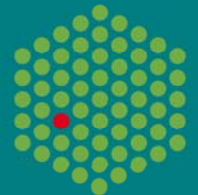


Data reduction and processing

Al Kikhney

BI((O))SAXS

EMBL

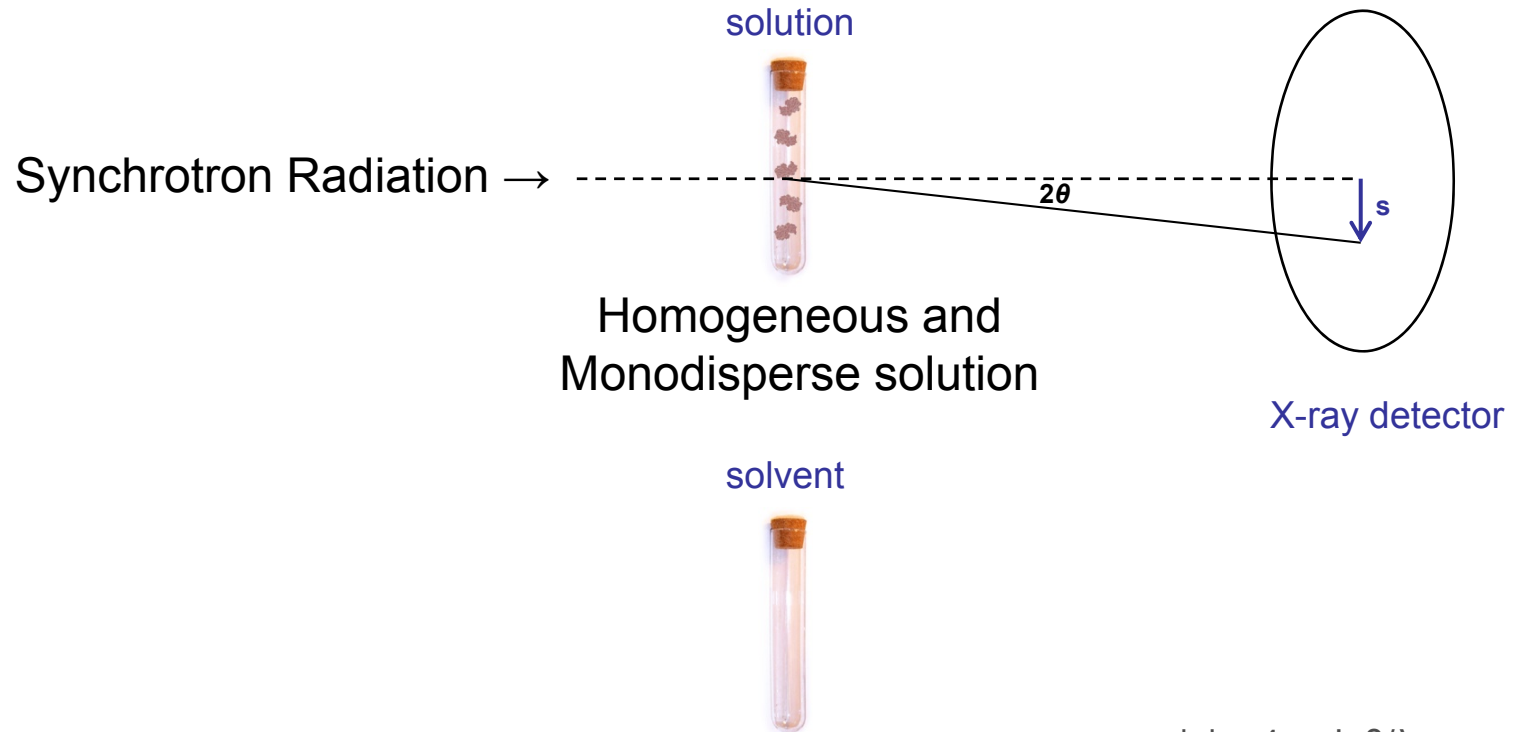


Outline

- SAXS experiment setup
- 3D $\rightarrow$ 2D $\rightarrow$ 1D
- Background subtraction
- High vs. low concentration
- R_g , MM
- Volume
- Distance distribution function $p(r)$



Small Angle X-ray Scattering



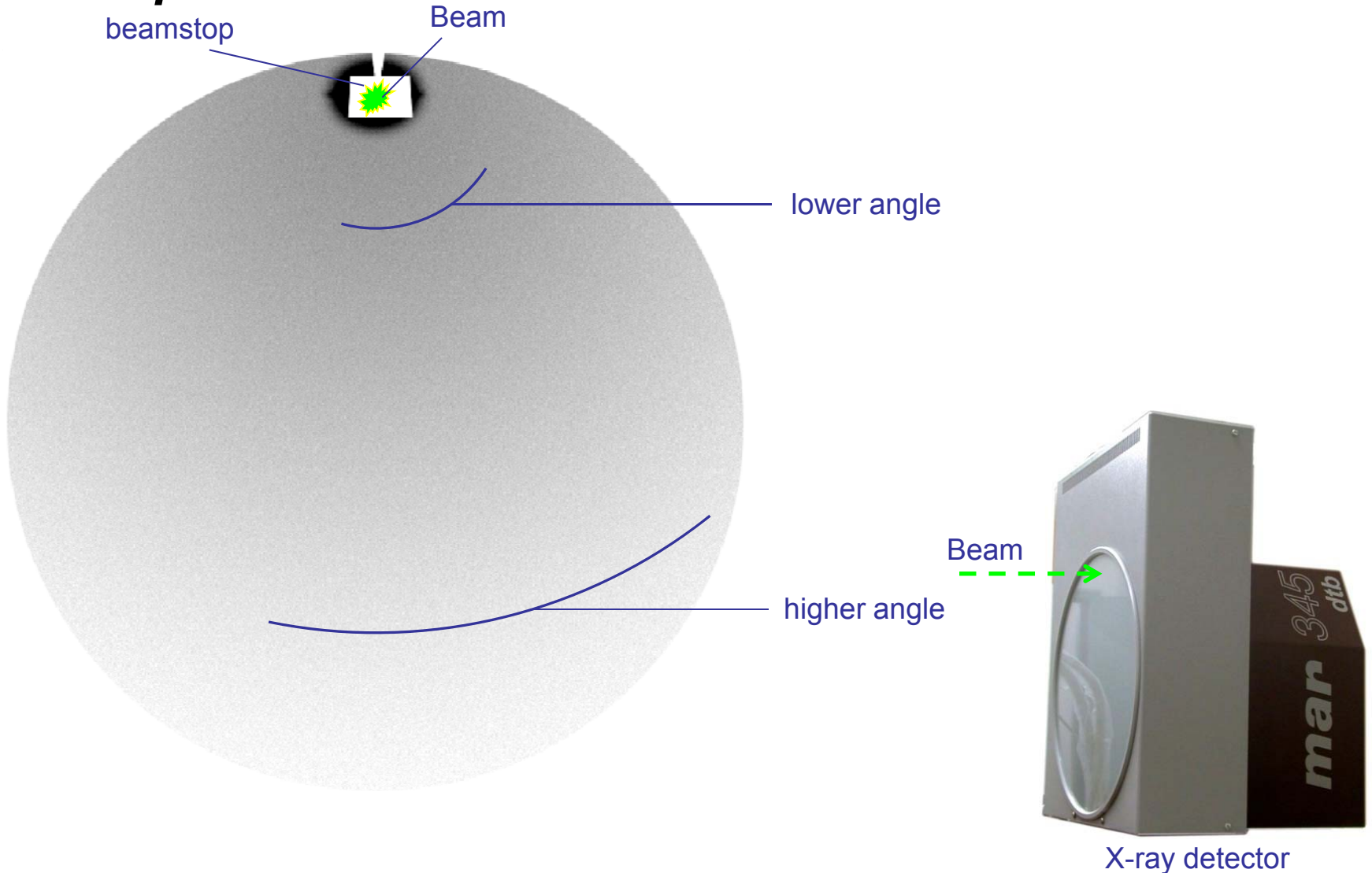
$$|s| = 4\pi \sin\theta/\lambda$$

1-2 mg purified material
concentration from 0.5 mg/ml,
exposure times: a few seconds/minutes

s – scattering vector
2θ – scattering angle
λ – wavelength
I(s) – intensity

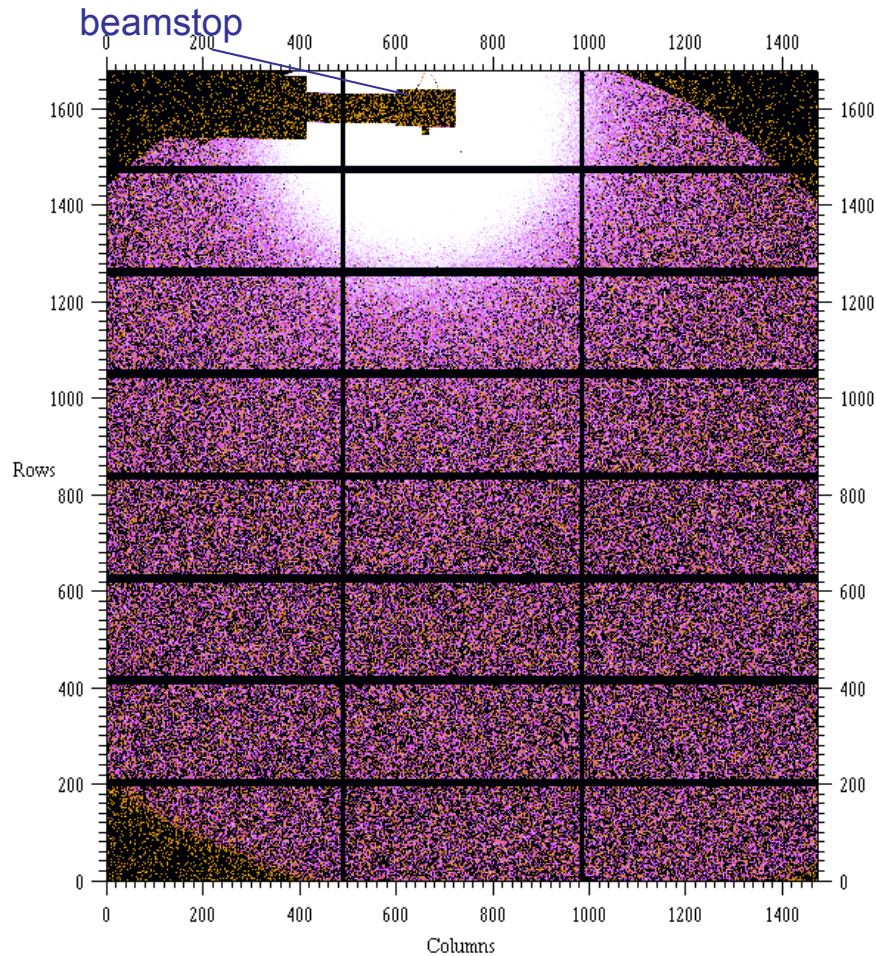
Small Angle X-ray Scattering

Exposure



Small Angle X-ray Scattering

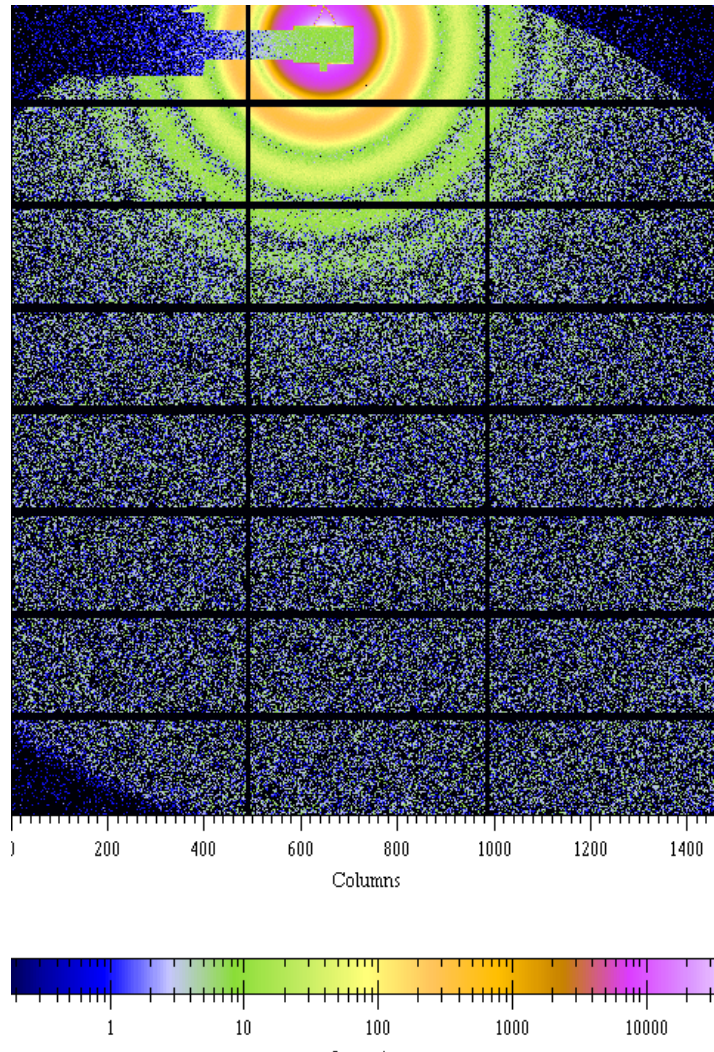
Exposure



X-ray detector

Small Angle X-ray Scattering

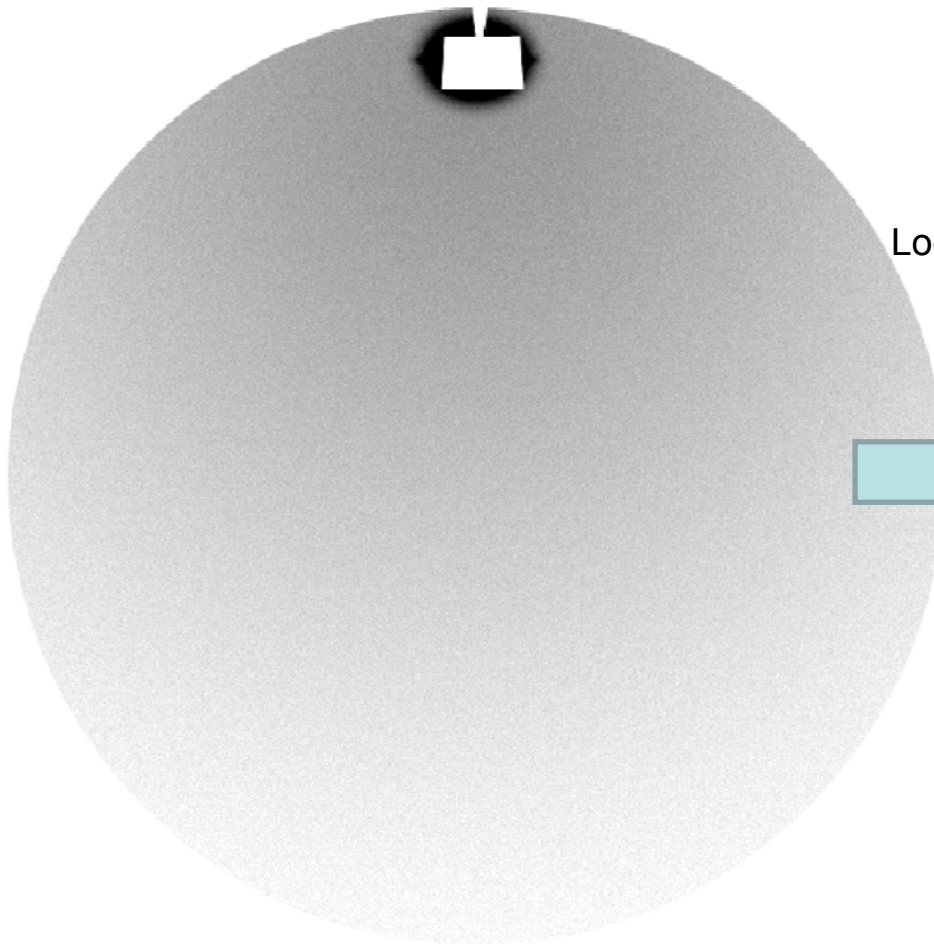
Exposure



X-ray detector

Small Angle X-ray Scattering

Radial averaging



Log $I(s)$,
a.u.

4.0

3.5

3.0

2.5

0

1.0

2.0

3.0

4.0

s , nm⁻¹

Normalization against:

- data collection time,
- transmitted sample intensity

Small Angle X-ray Scattering

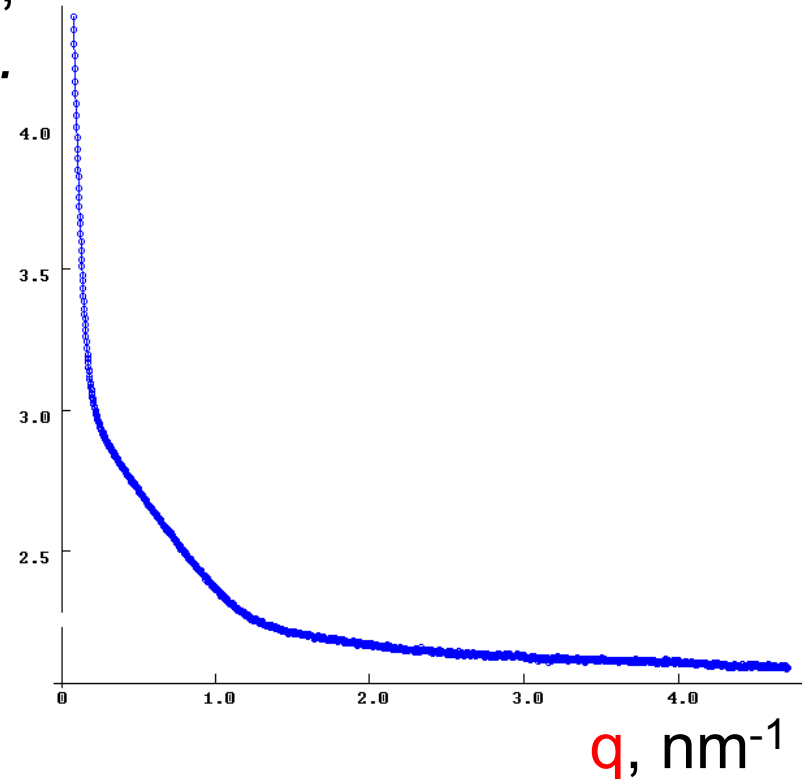
Notations and units

$$|q| = 4\pi \sin\theta/\lambda$$

2θ – scattering angle

λ – wavelength

Log $I(q)$,
a.u.



Small Angle X-ray Scattering

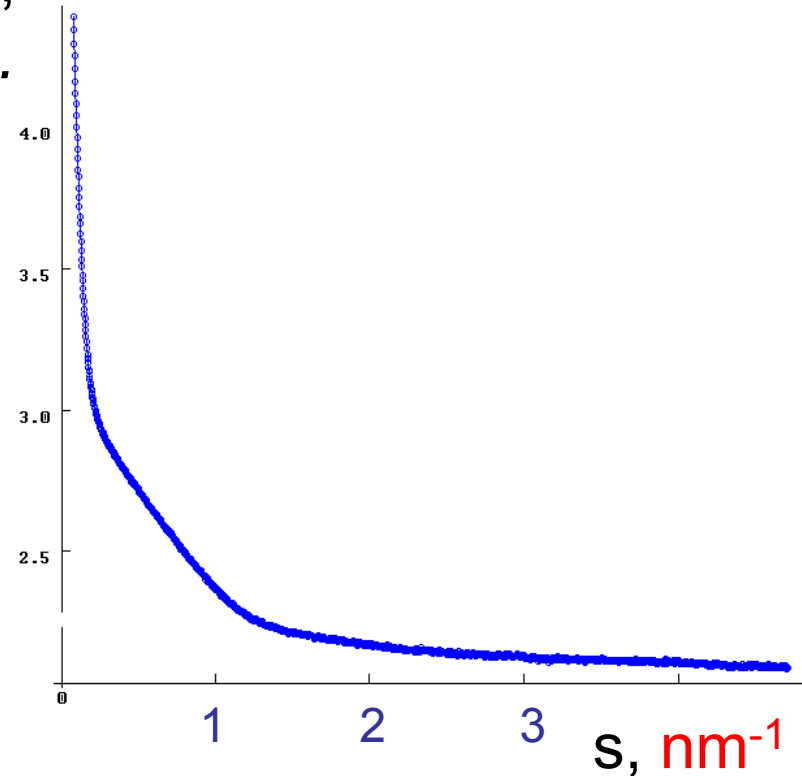
Notations and units

$$|s| = 4\pi \sin\theta/\lambda$$

2θ – scattering angle

λ – wavelength

Log $I(s)$,
a.u.



Small Angle X-ray Scattering

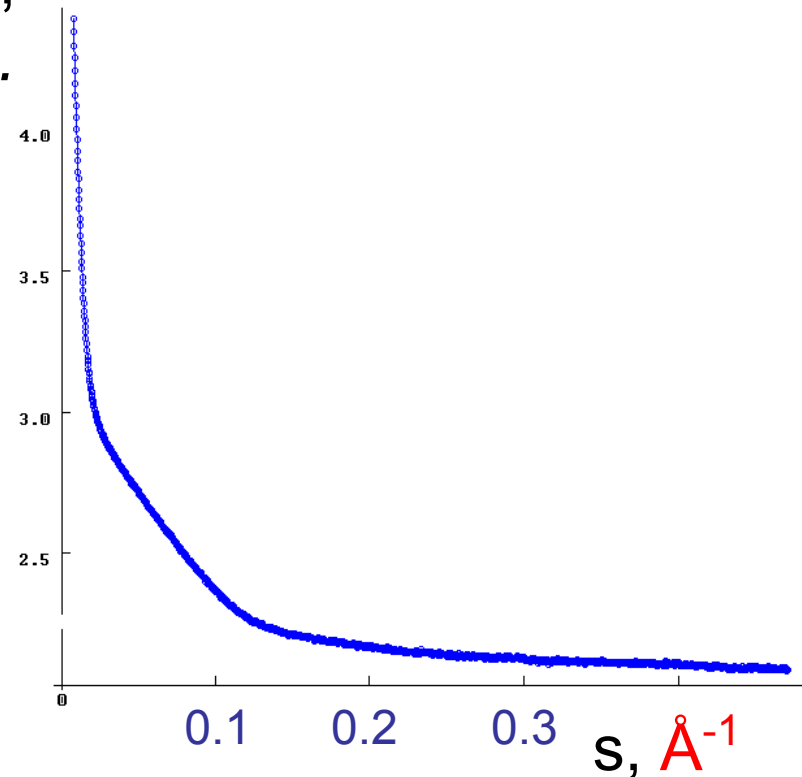
Notations and units

$$|s| = 4\pi \sin\theta/\lambda$$

2θ – scattering angle

λ – wavelength

Log $I(s)$,
a.u.



Small Angle X-ray Scattering

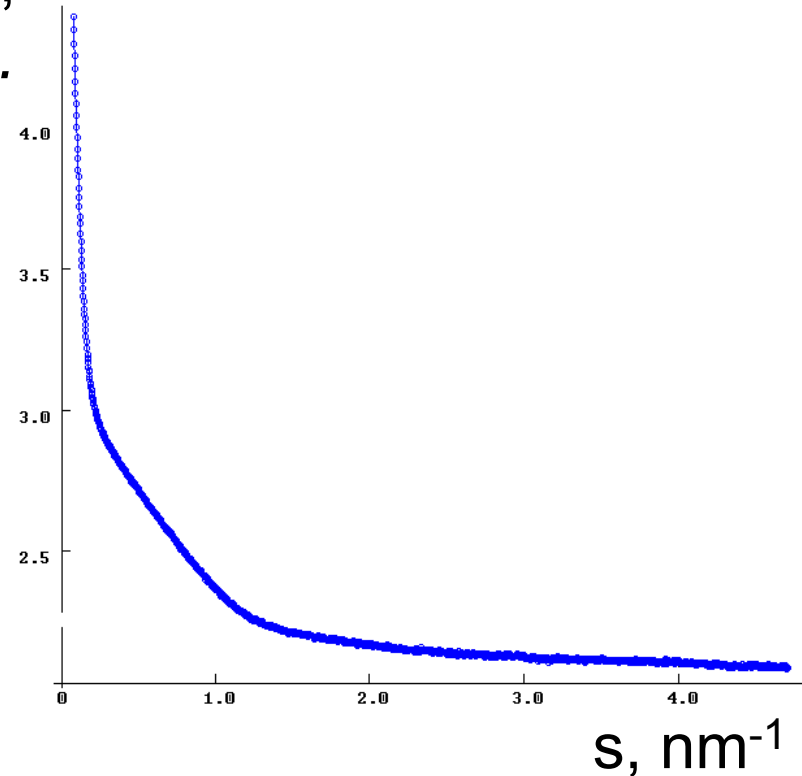
Notations and units

$$|s| = 2 \sin\theta/\lambda$$

2θ – scattering angle

λ – wavelength

Log $I(s)$,
a.u.



Small Angle X-ray Scattering

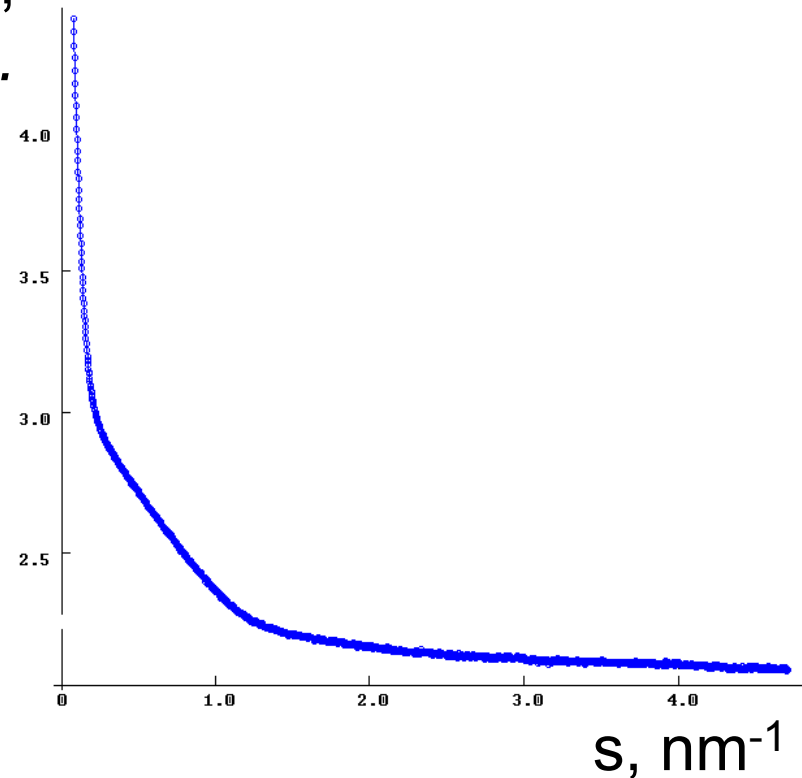
Notations and units

$$|s| = 4\pi \sin\theta/\lambda$$

2θ – scattering angle

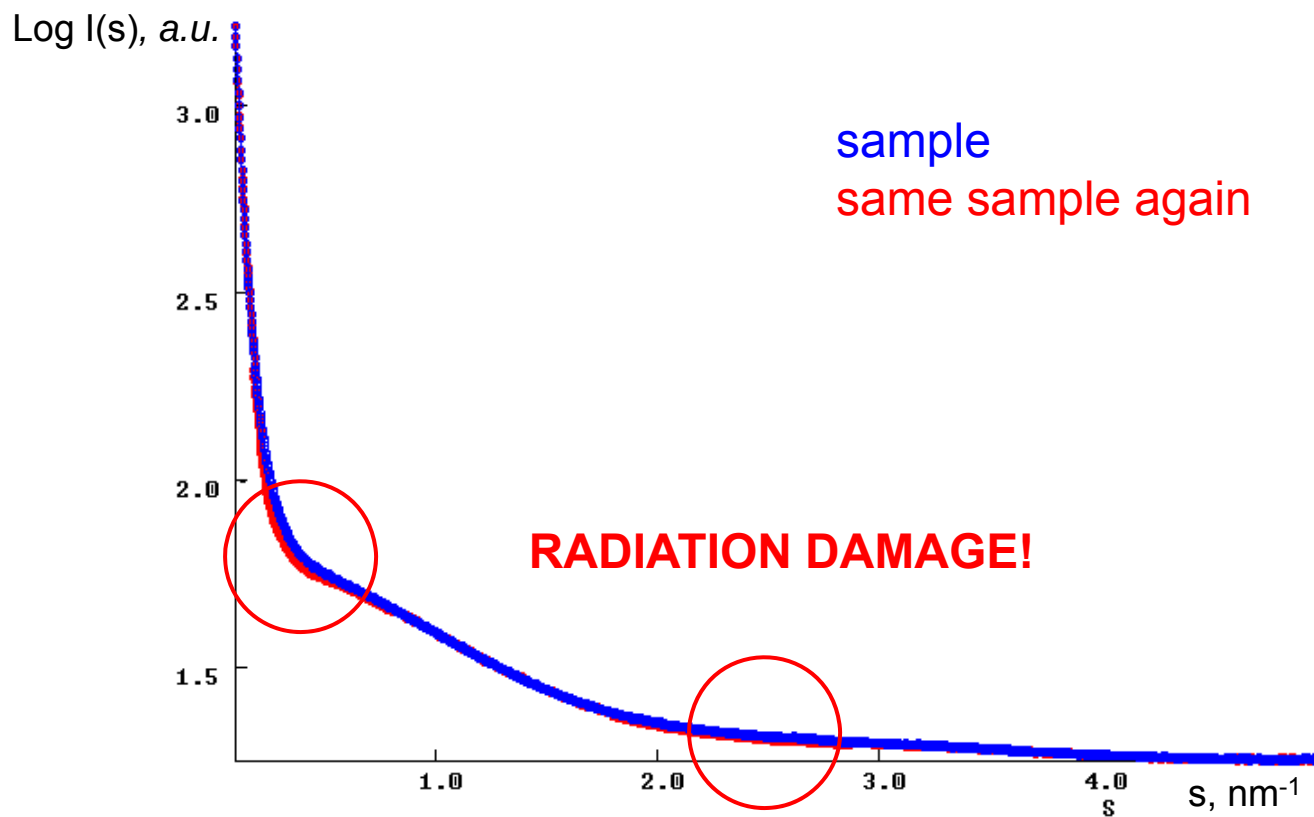
λ – wavelength

Log $I(s)$,
a.u.

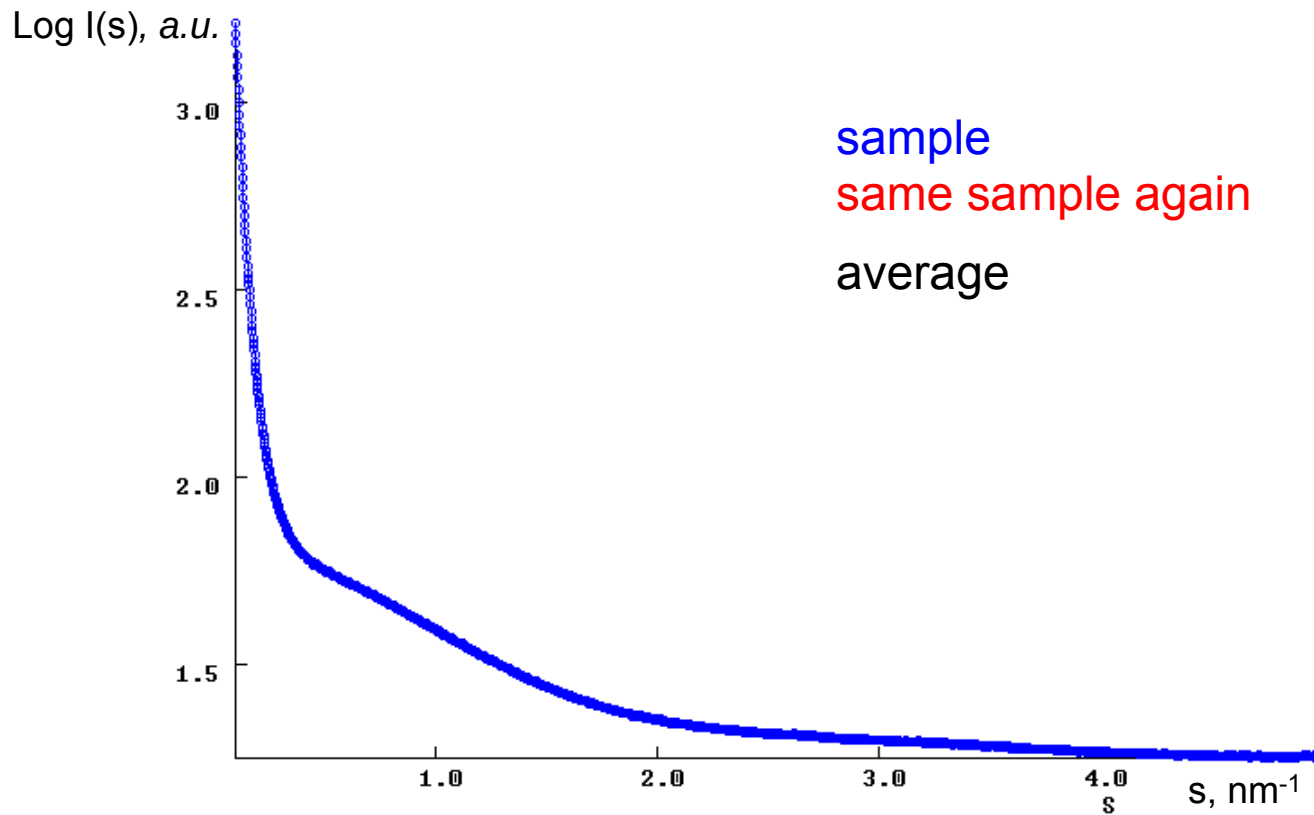


Data quality

Radiation damage

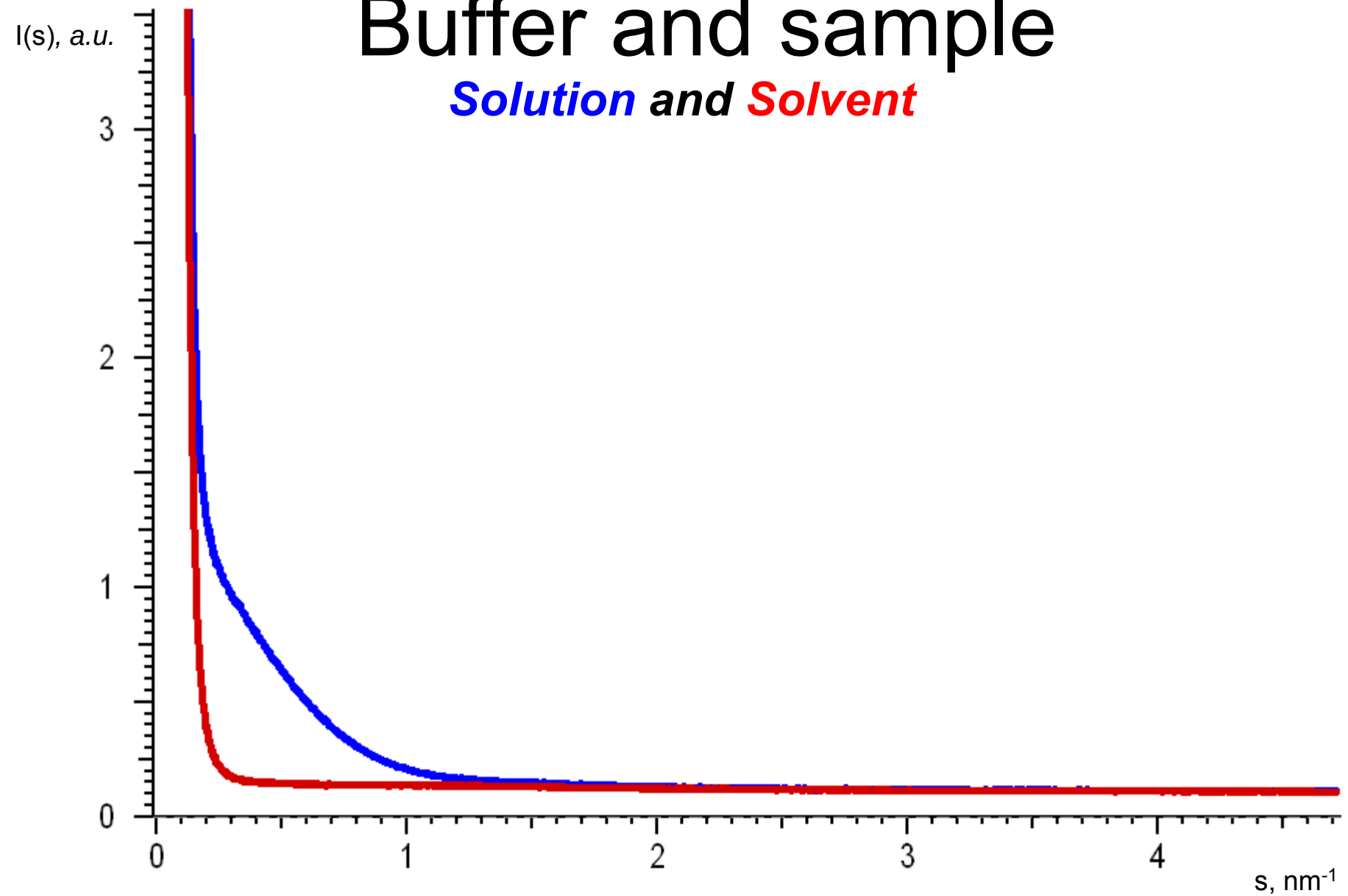


Data quality



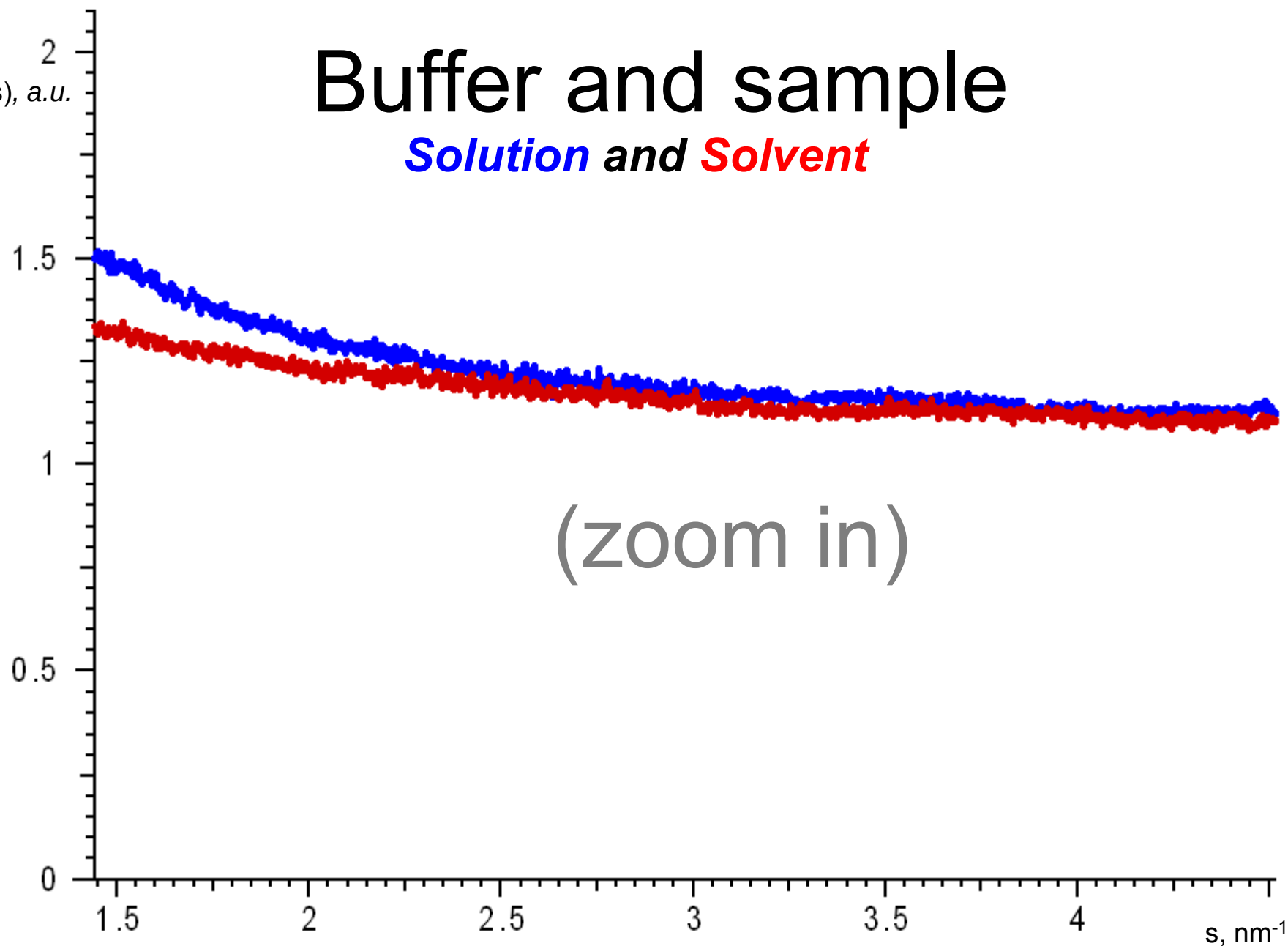
Buffer and sample

Solution and Solvent



Buffer and sample

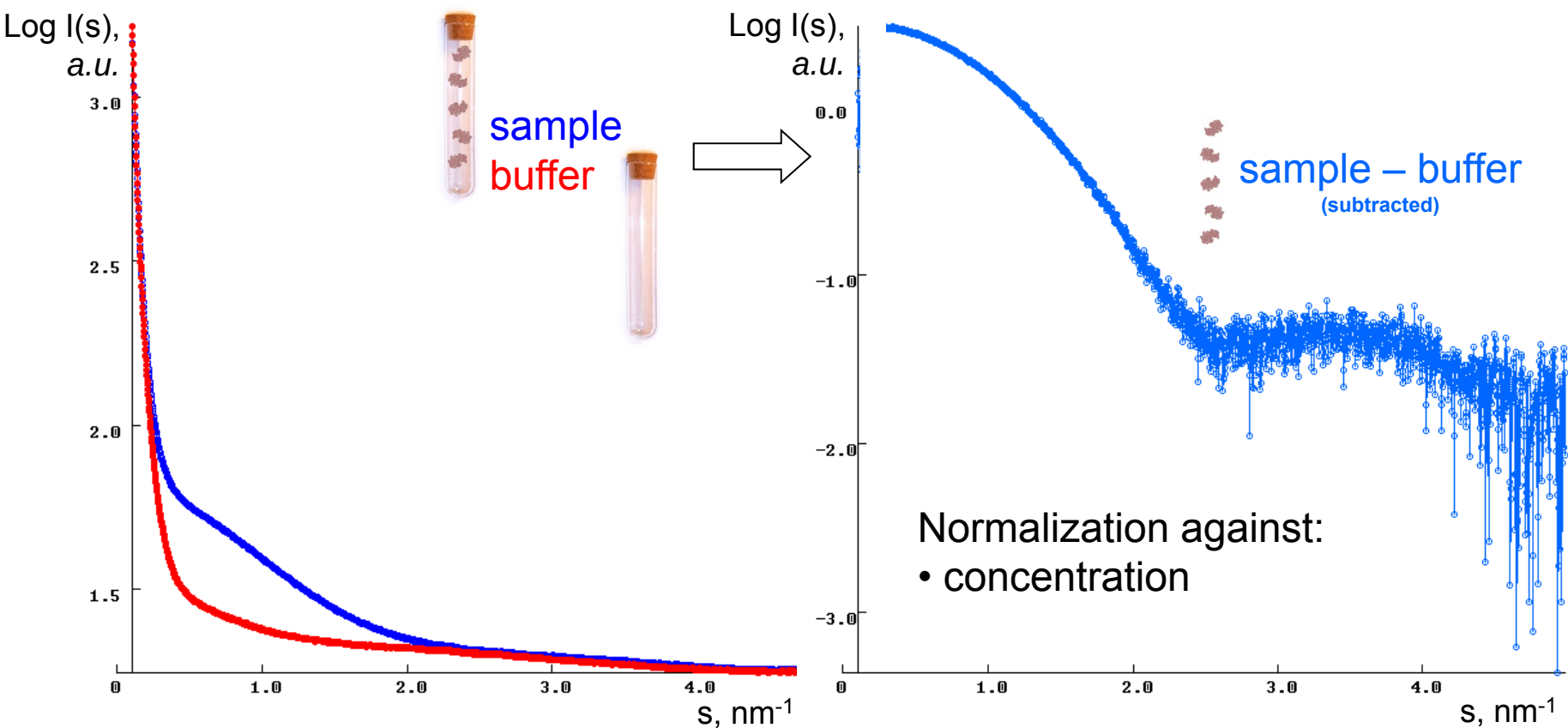
Solution and Solvent



Looking for protein signals less than 5% above background level...

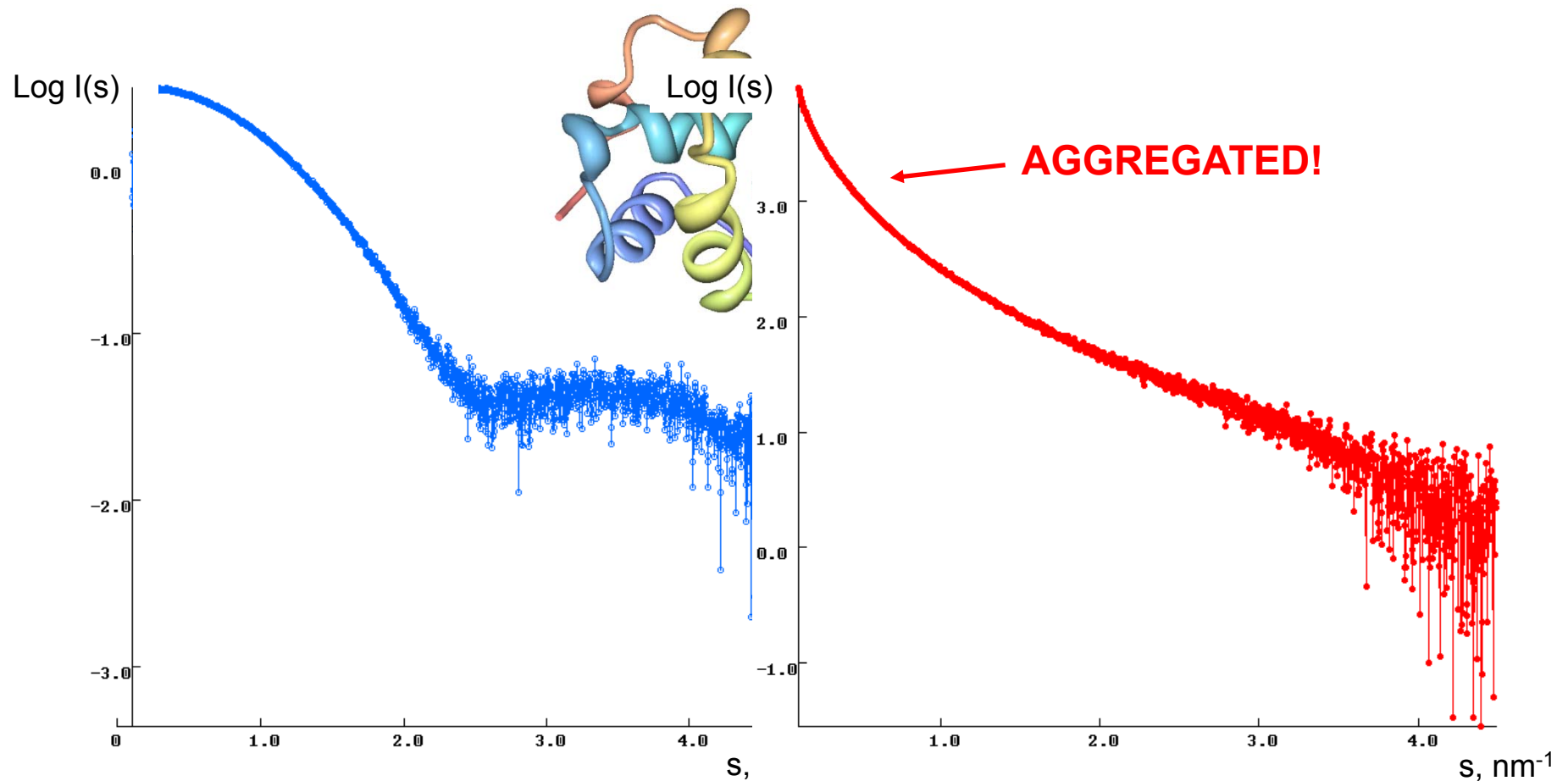
Background subtraction

Solution minus Solvent



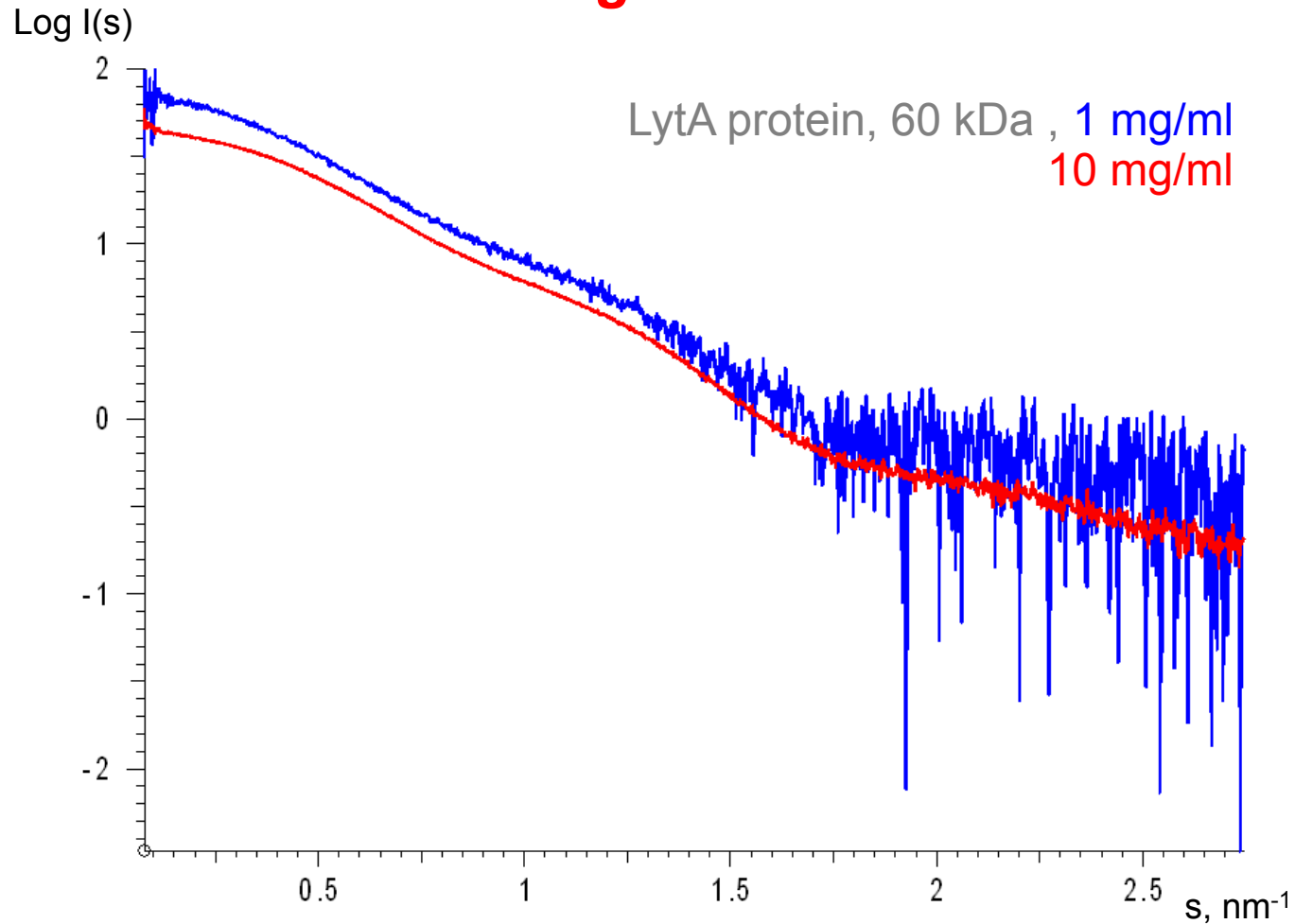
Data quality

“Can I use this data for further analysis?”



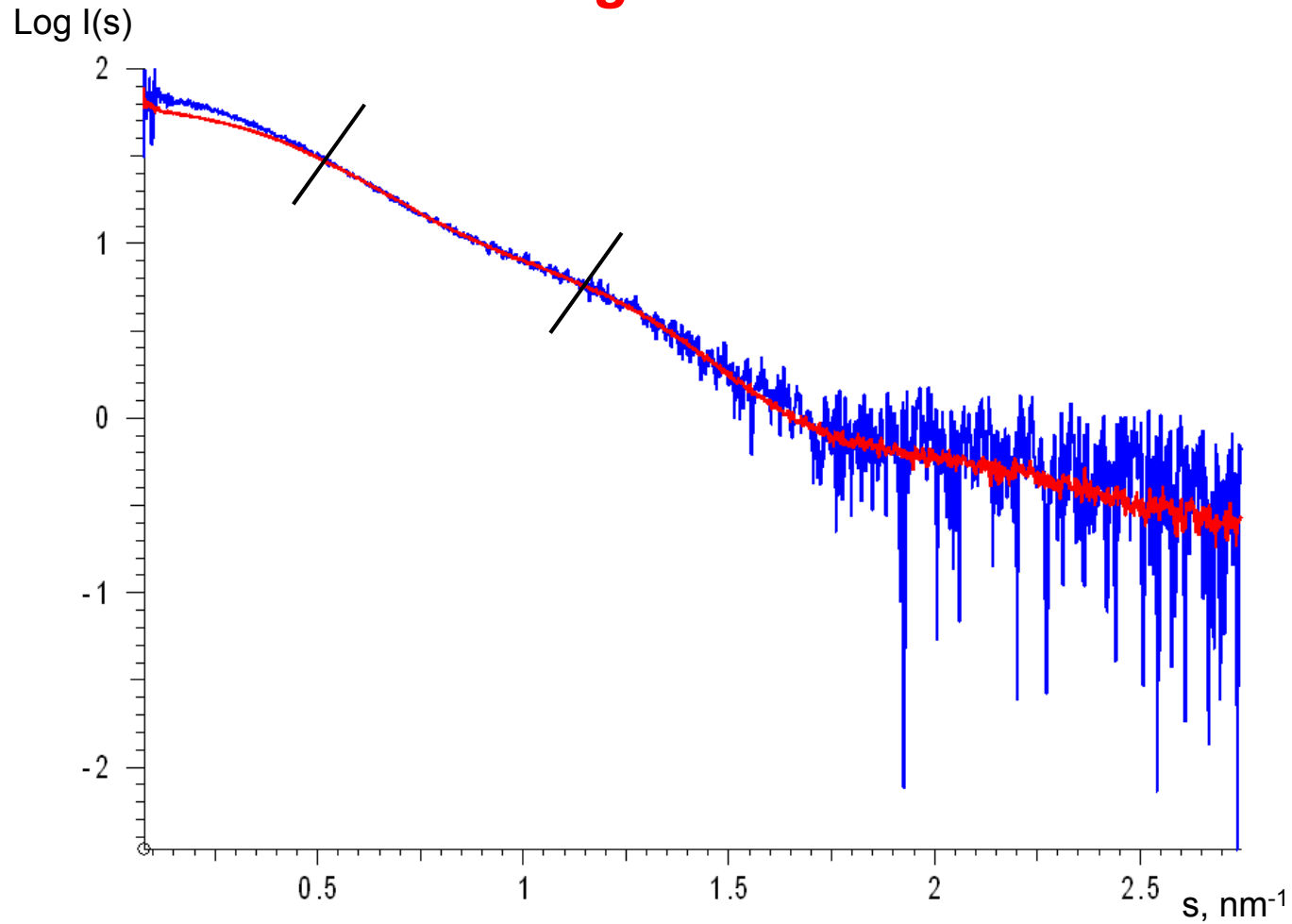
Dilution series

Low and High Concentration



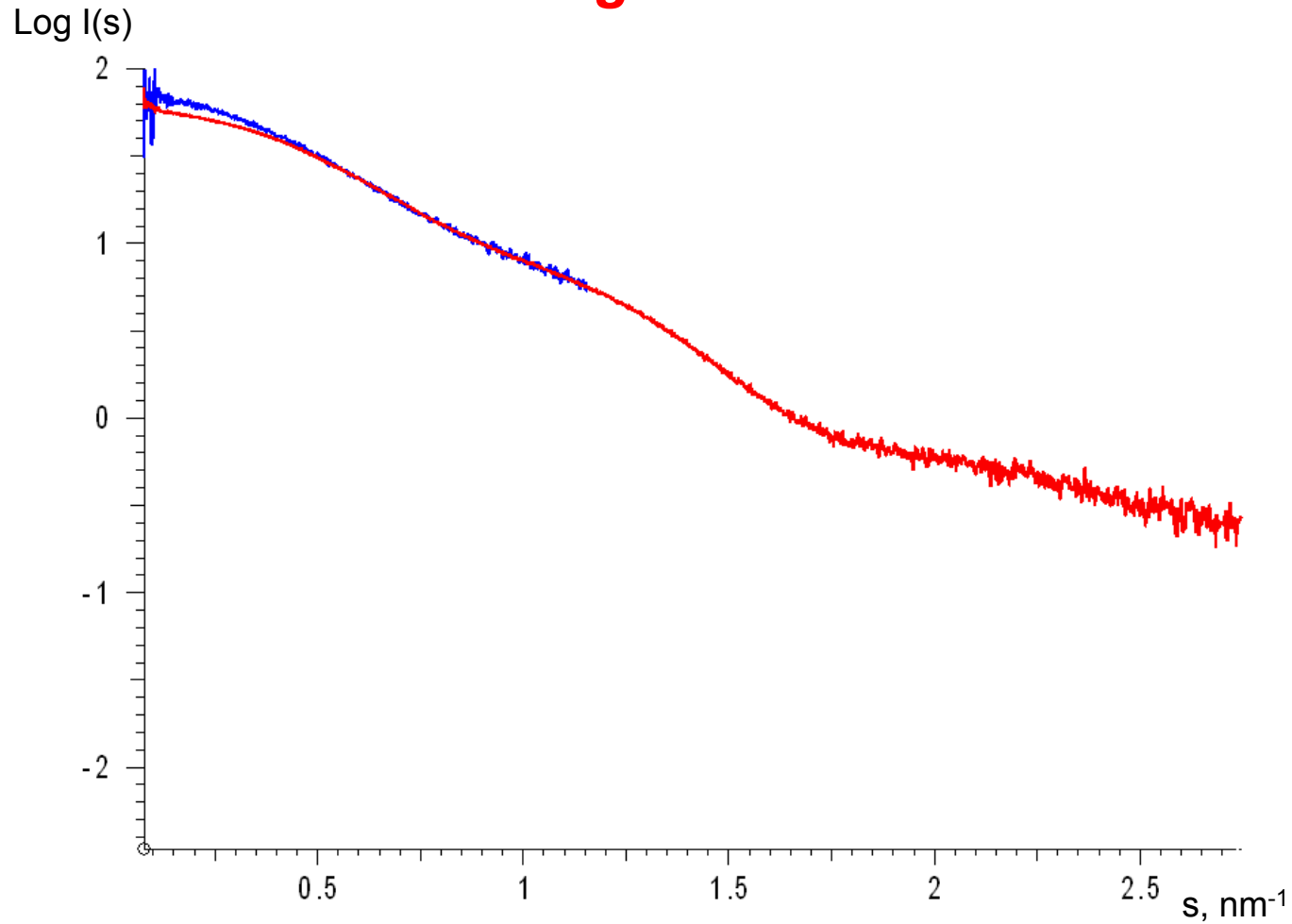
Dilution series

Low and High Concentration



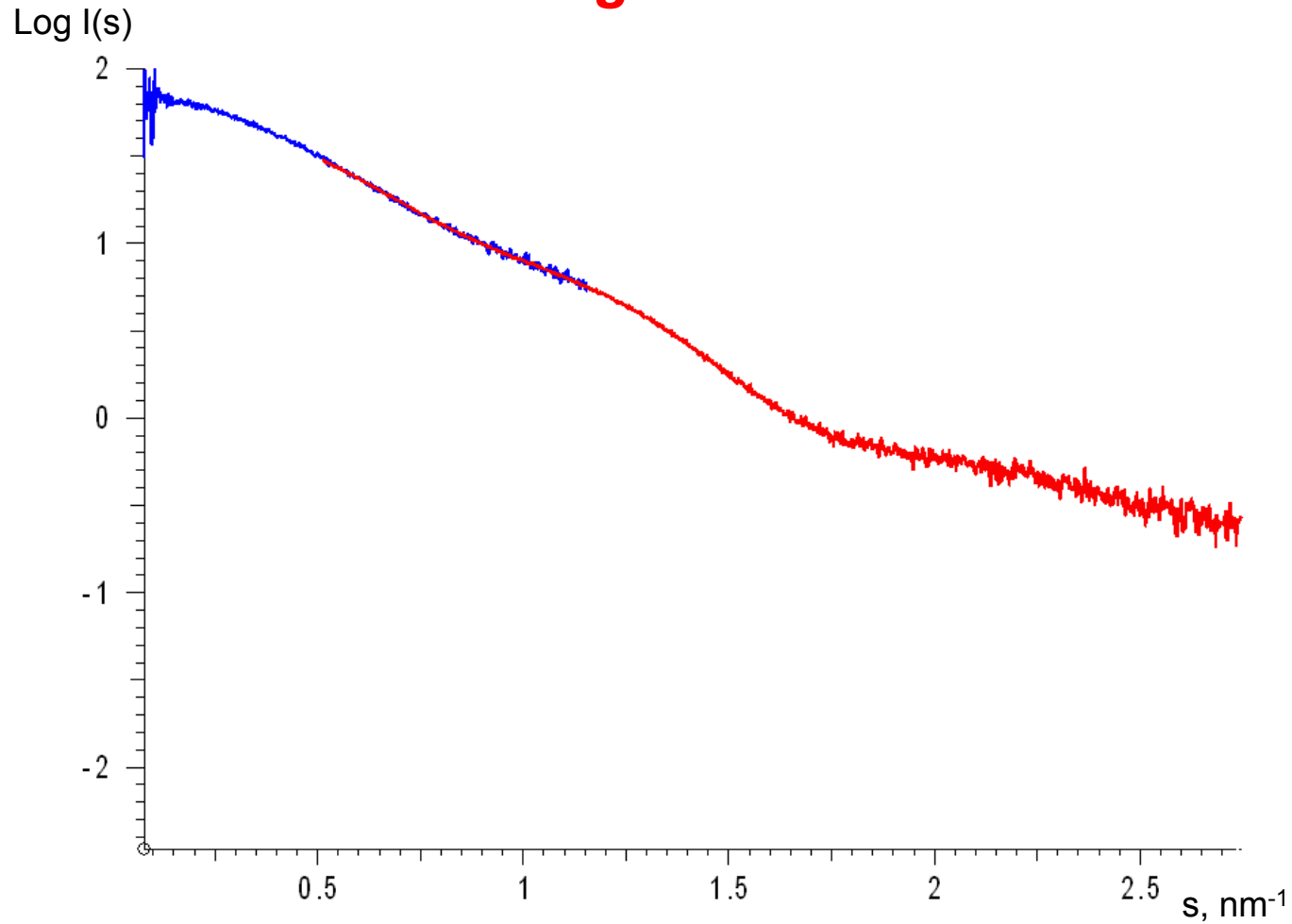
Merging data

Low and High Concentration



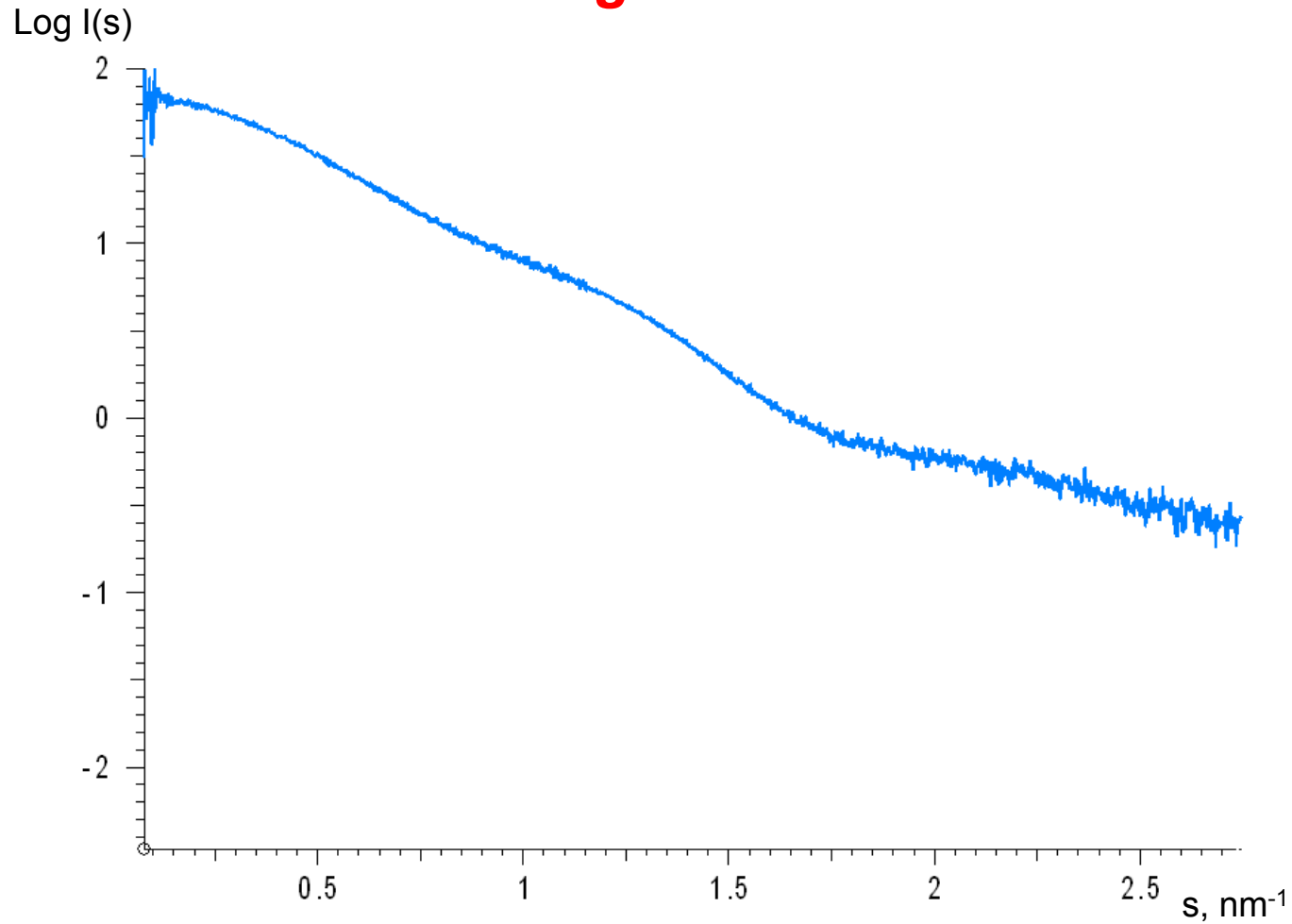
Merging data

Low and High Concentration



Merging data

Low and High Concentration



Data analysis

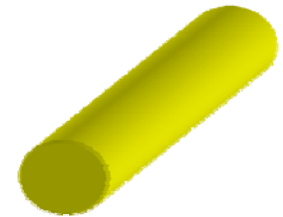
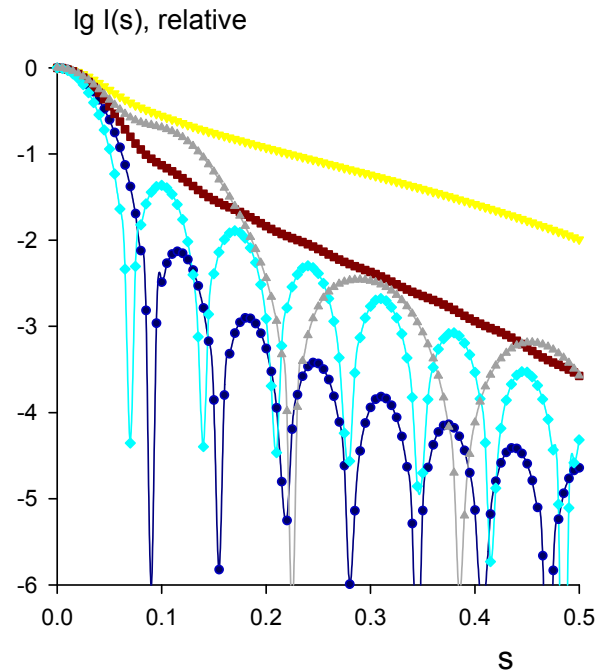
Shape



Solid sphere



Hollow sphere



Long rod

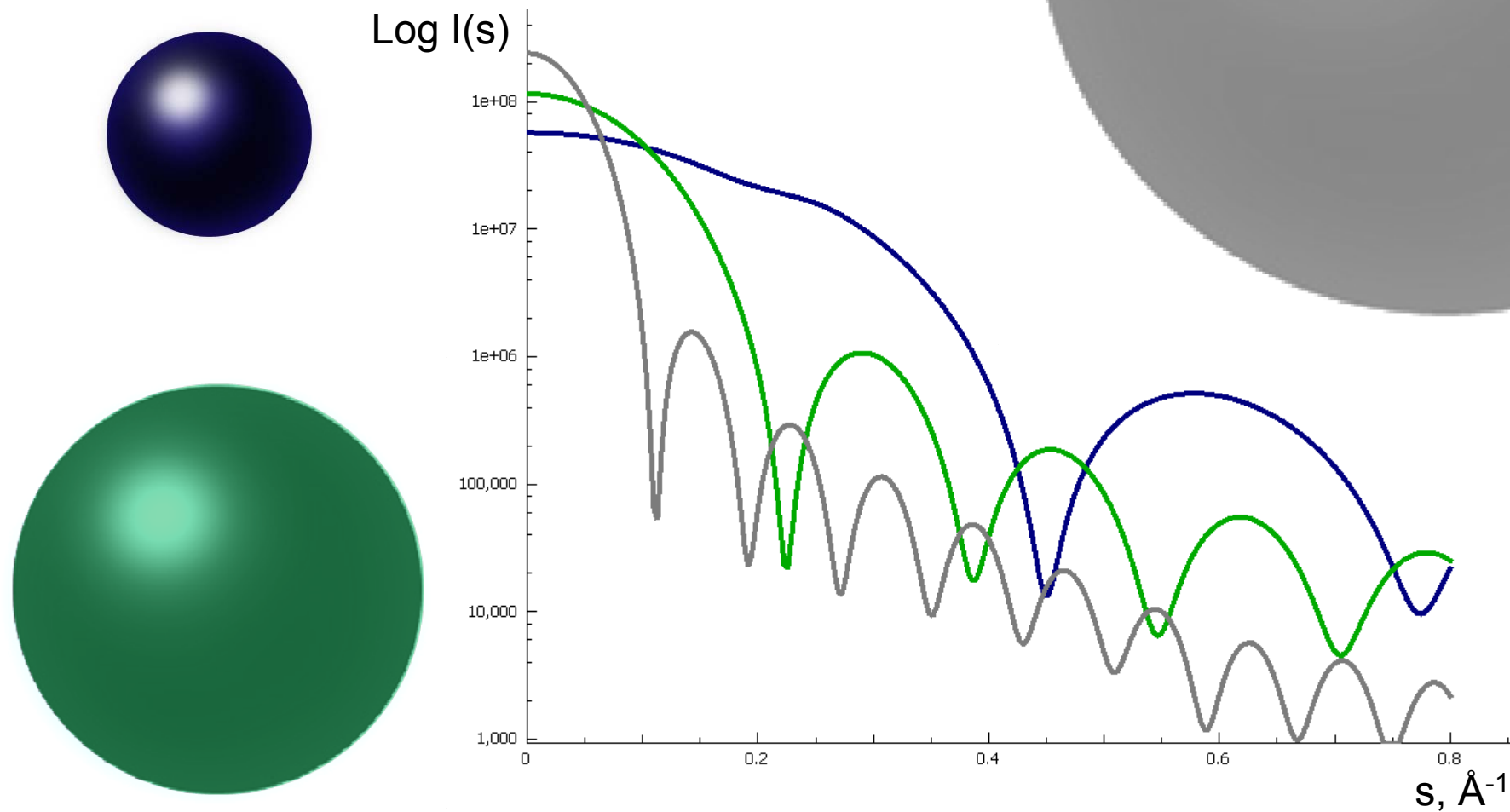


Flat disc

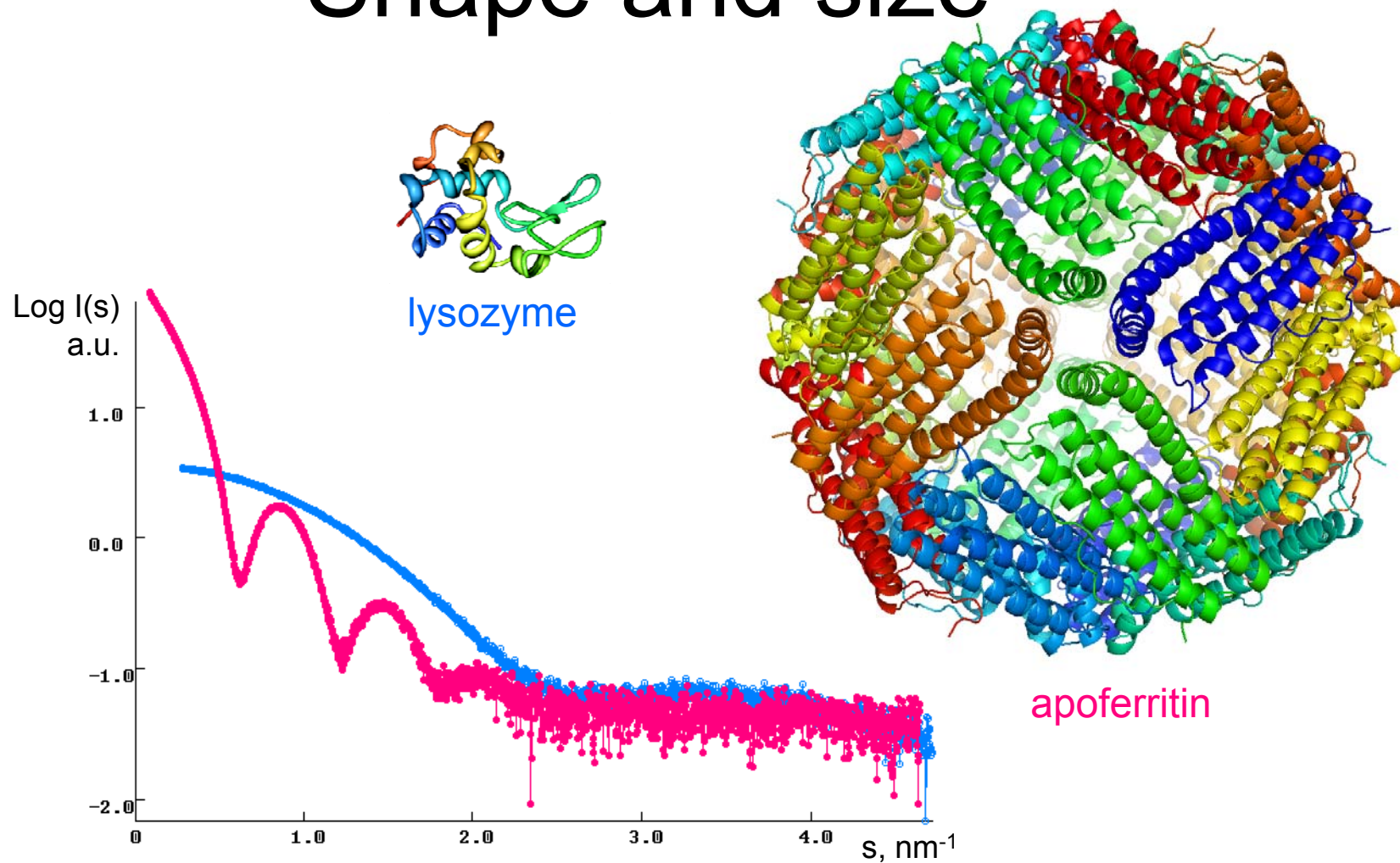


Dumbbell

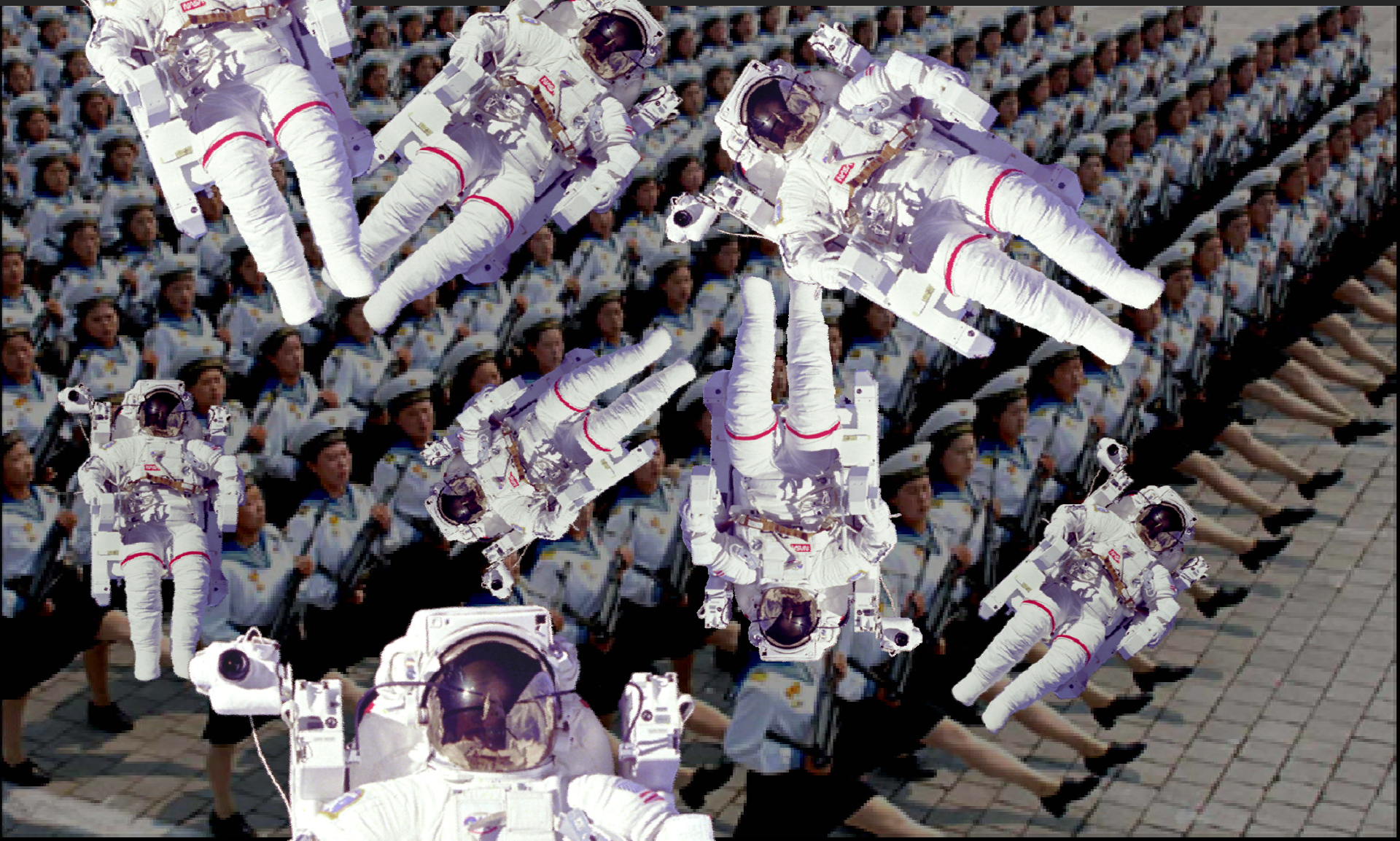
Size



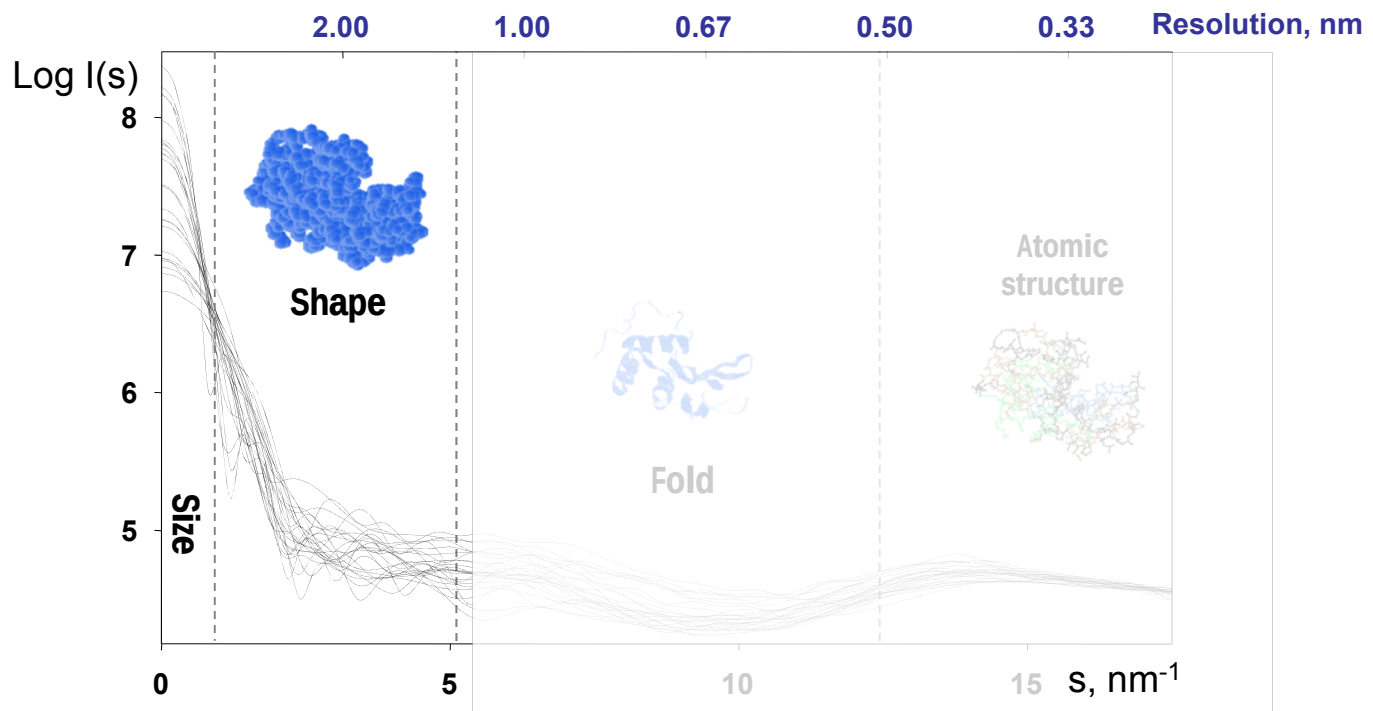
Shape and size



Crystal vs. solution



Data range



Radius of gyration (R_g)

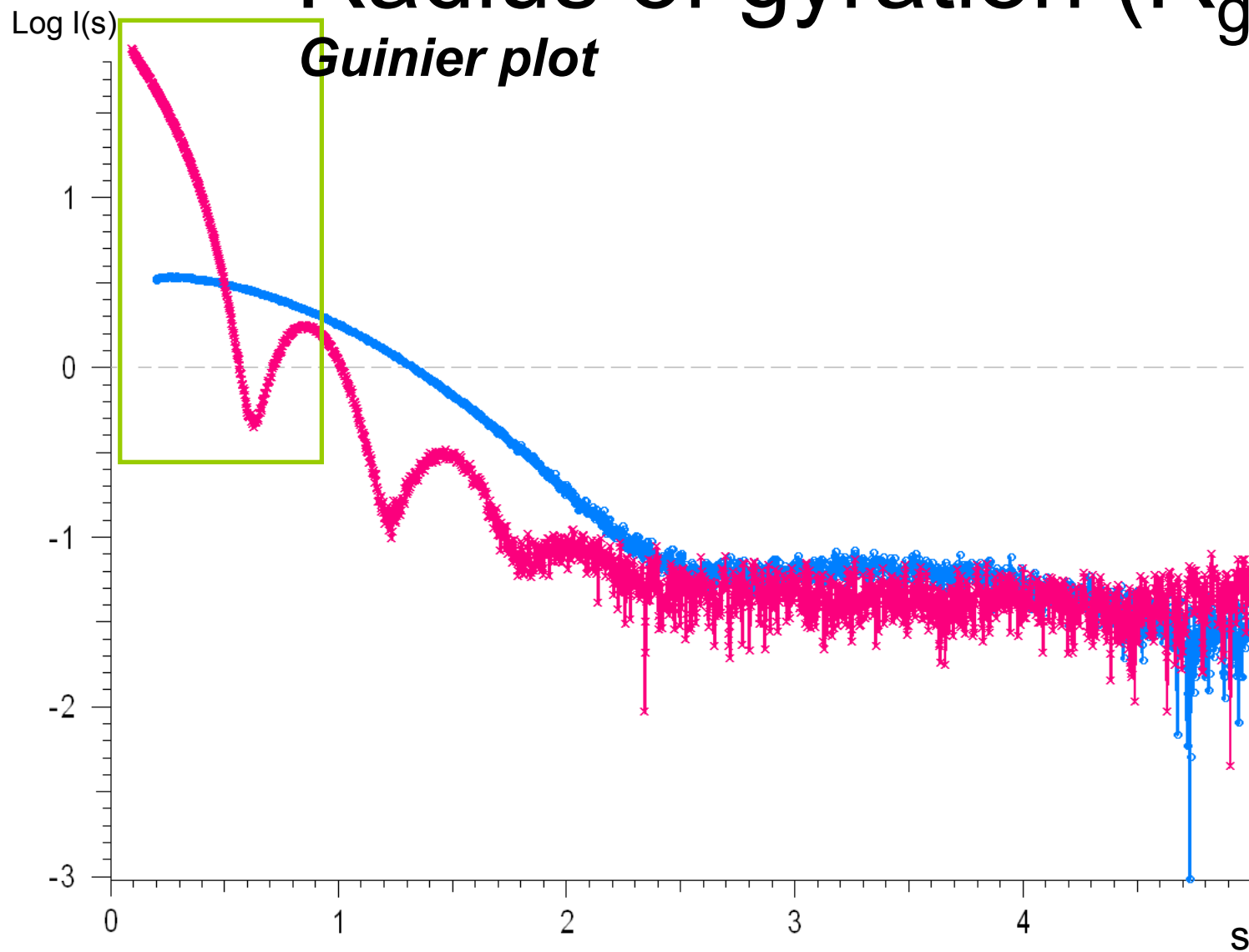
Definition

Measure for the overall size of a macromolecule

Average of square center-of-mass
distances in the molecule
weighted by the scattering length density

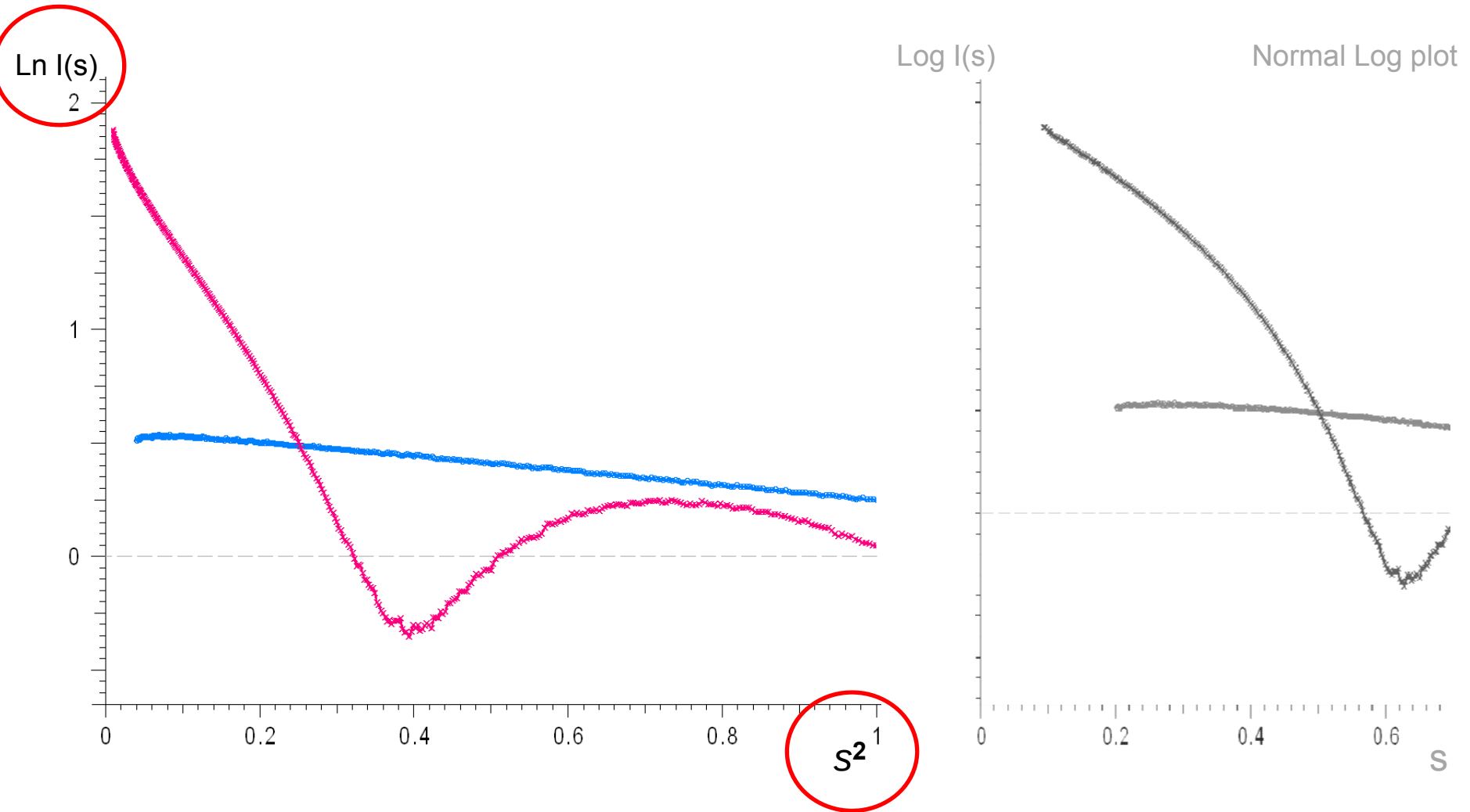
Radius of gyration (R_g)

Guinier plot



Radius of gyration (R_g)

Guinier plot



Radius of gyration (R_g)

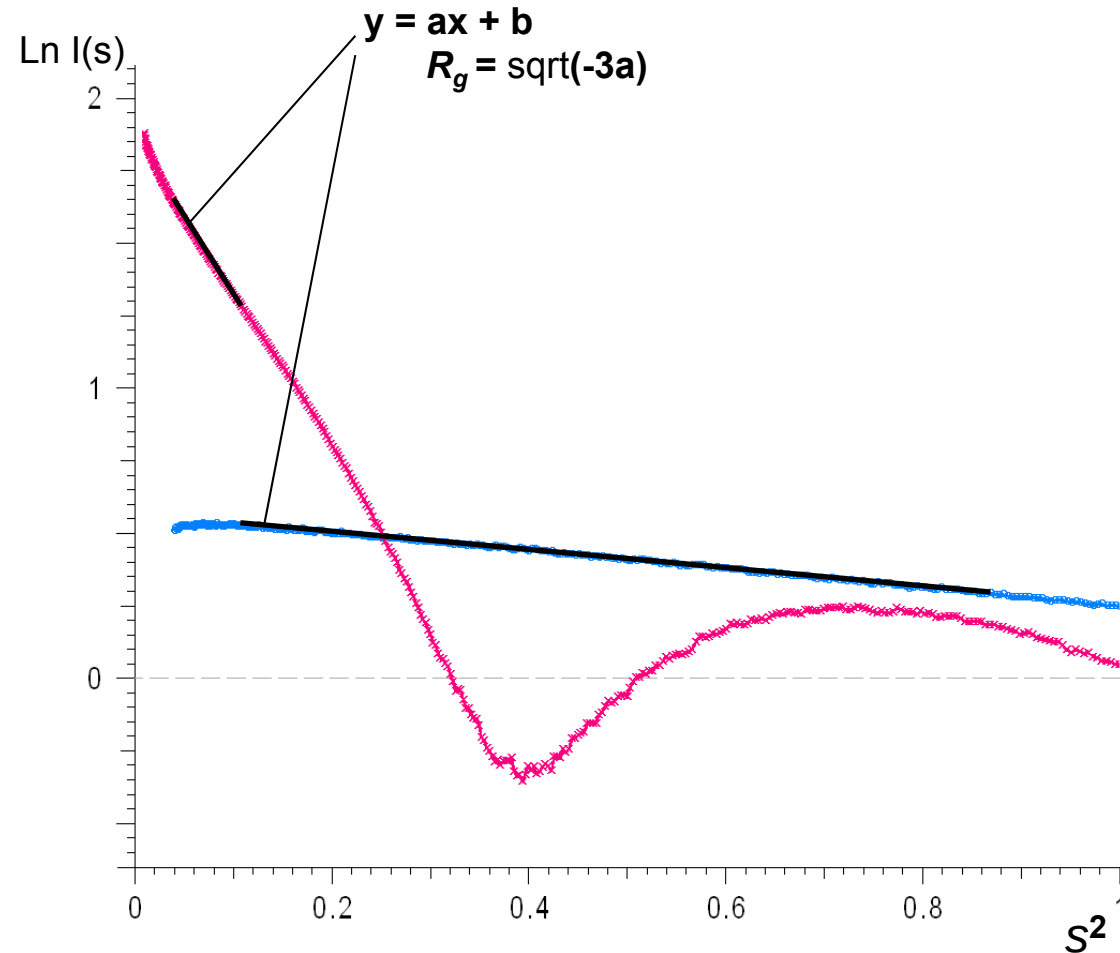
Guinier plot

- Estimate of the overall size of the particles

Guinier approximation:

$$I(s) = I(0)\exp(-s^2 R_g^2/3)$$

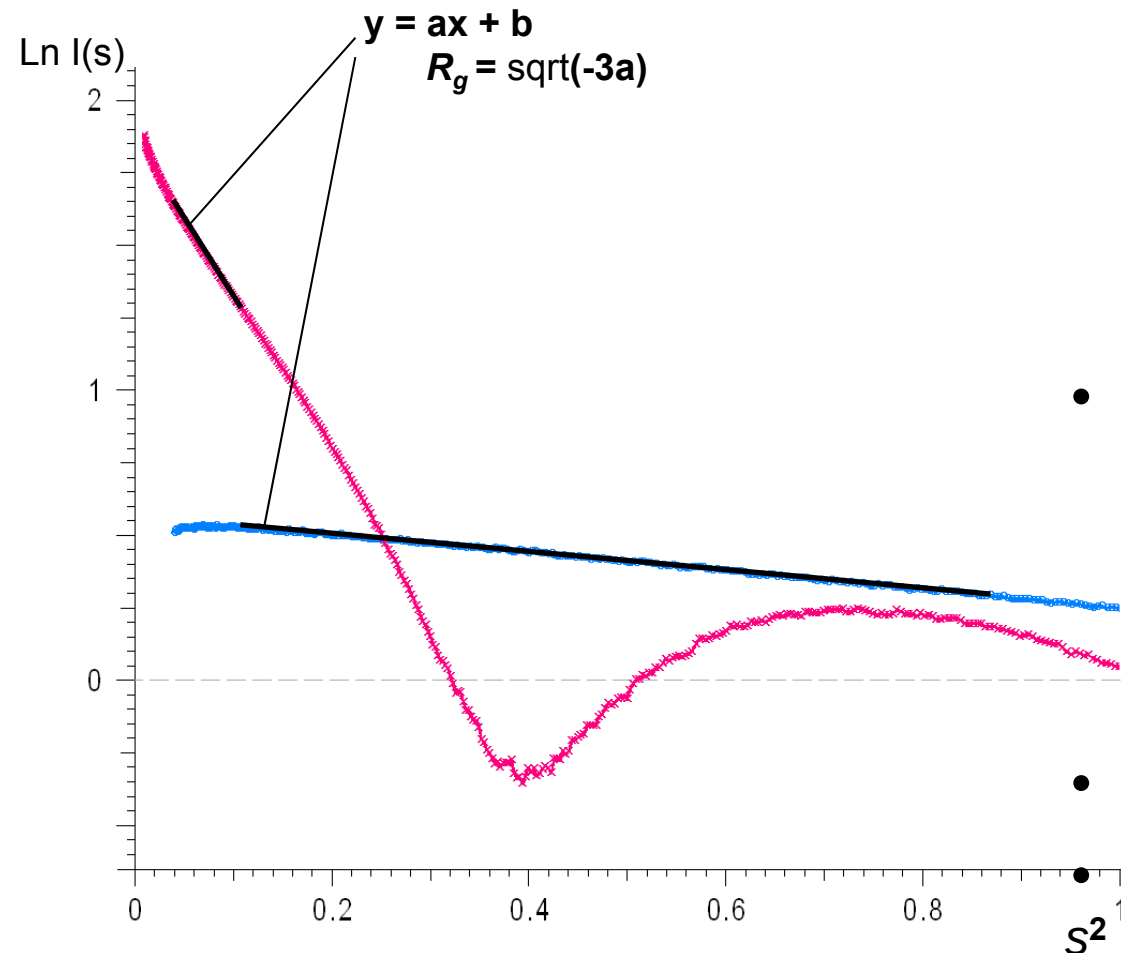
$$sR_g \lesssim 1.3$$



André Guinier
1911-2000

Radius of gyration (R_g)

Guinier plot



- Estimate of the overall size of the particles

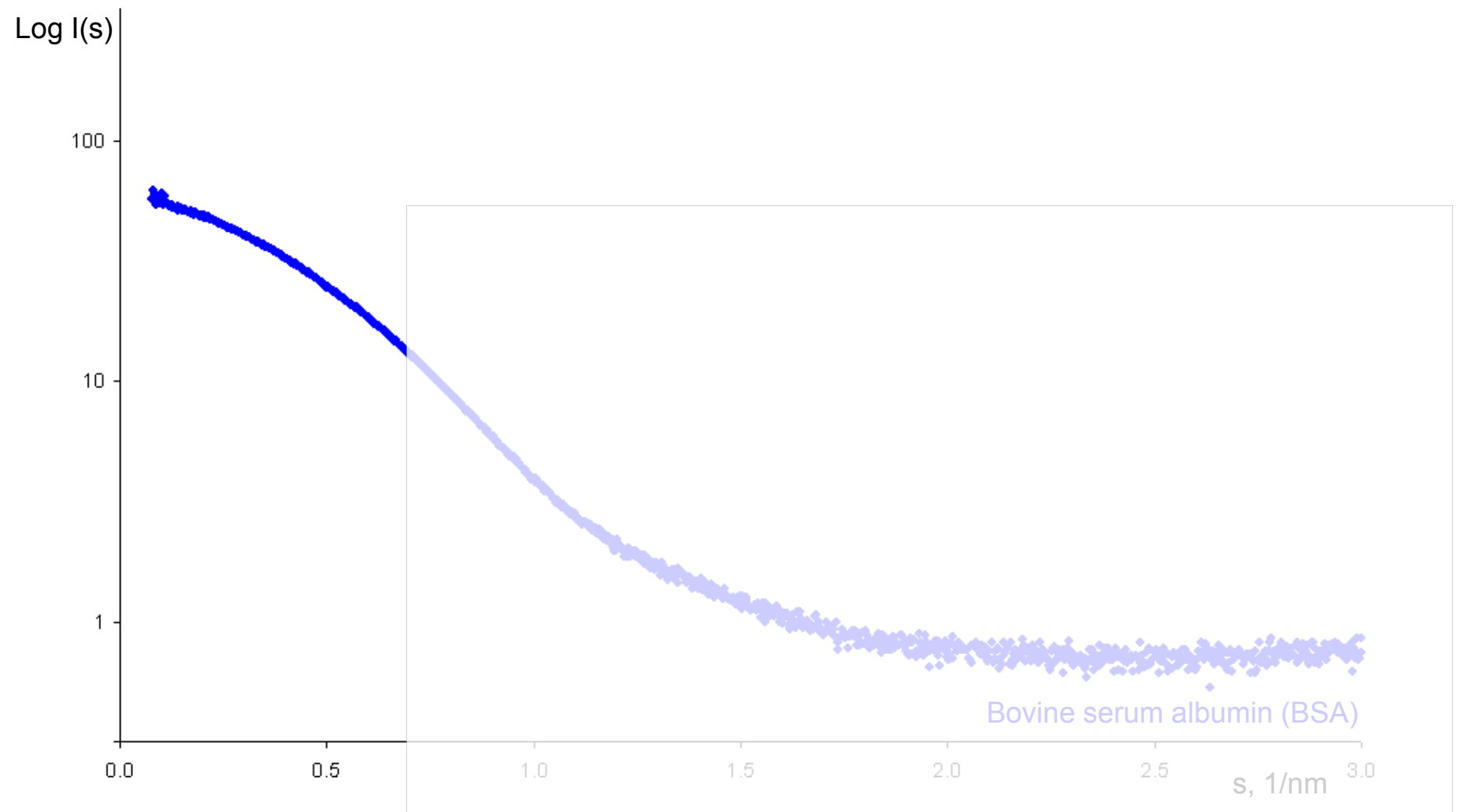
Guinier approximation:

$$I(s) = I(0)\exp(-s^2 R_g^2/3)$$

$$sR_g \lesssim 1.3$$

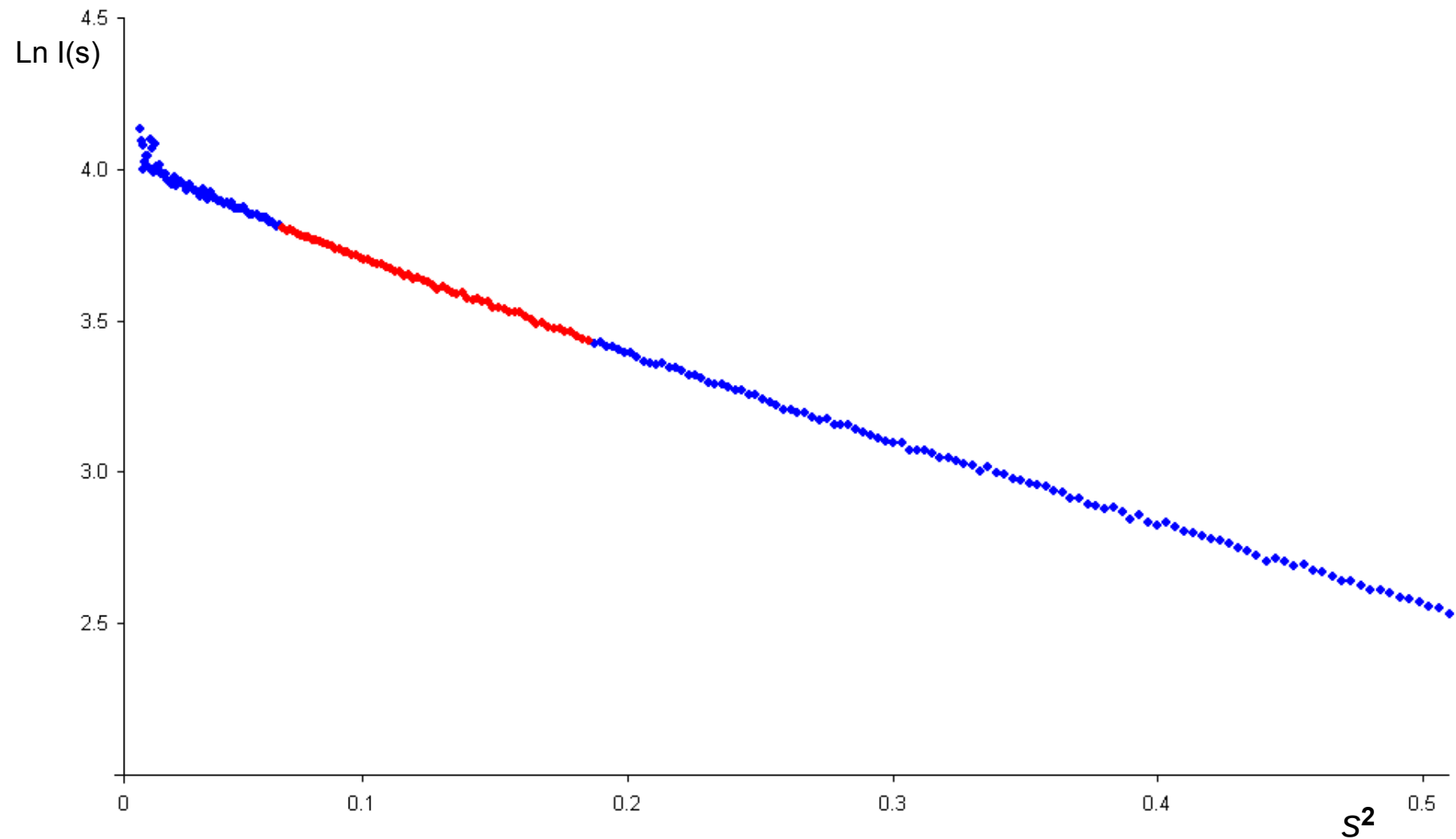
- Quality of the data
 - aggregation
 - polydispersity
 - improper background subtraction
- Zero angle intensity $I(0)$
- First point to use

Radius of gyration (R_g)



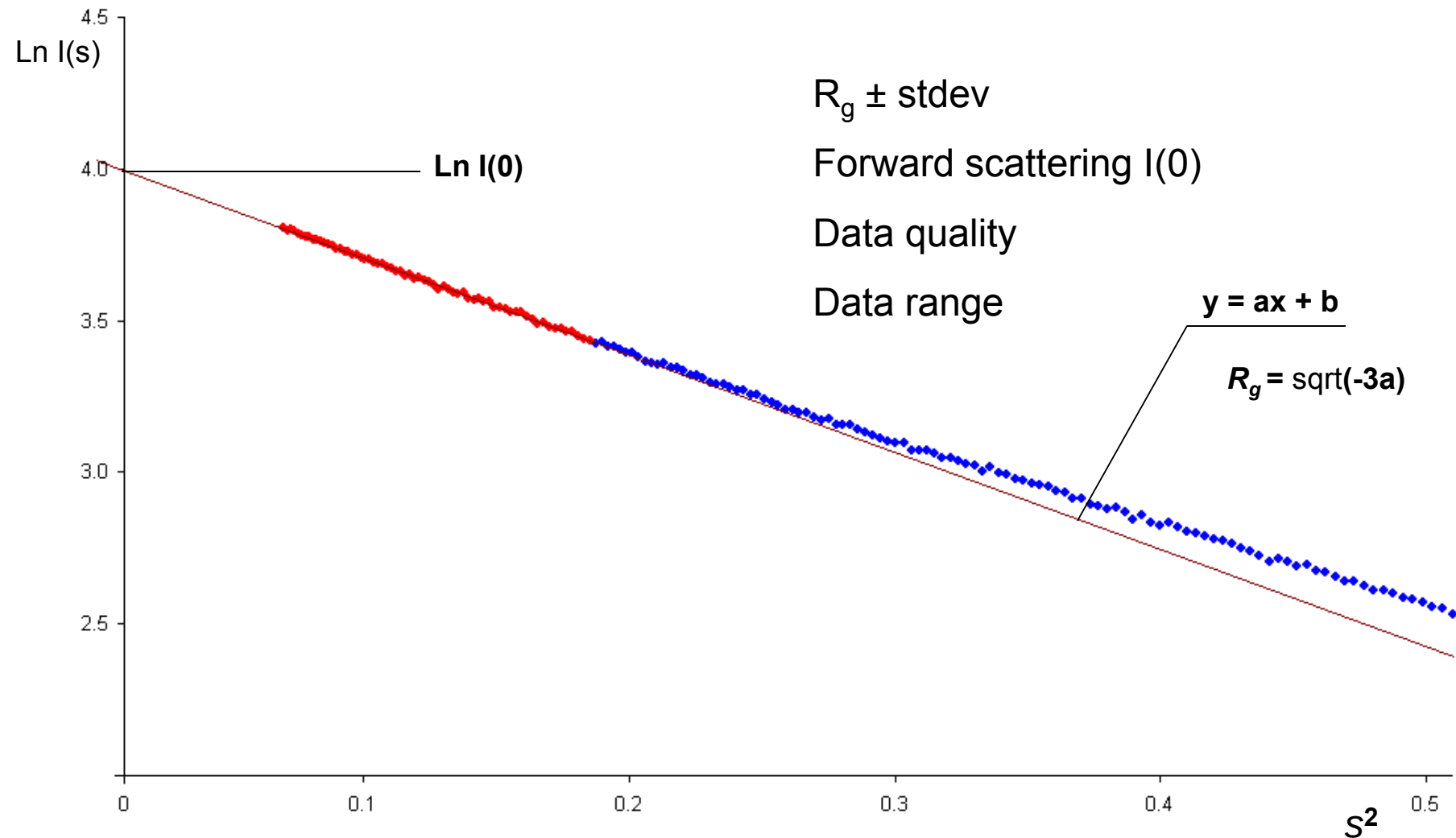
Radius of gyration (R_g)

Guinier plot



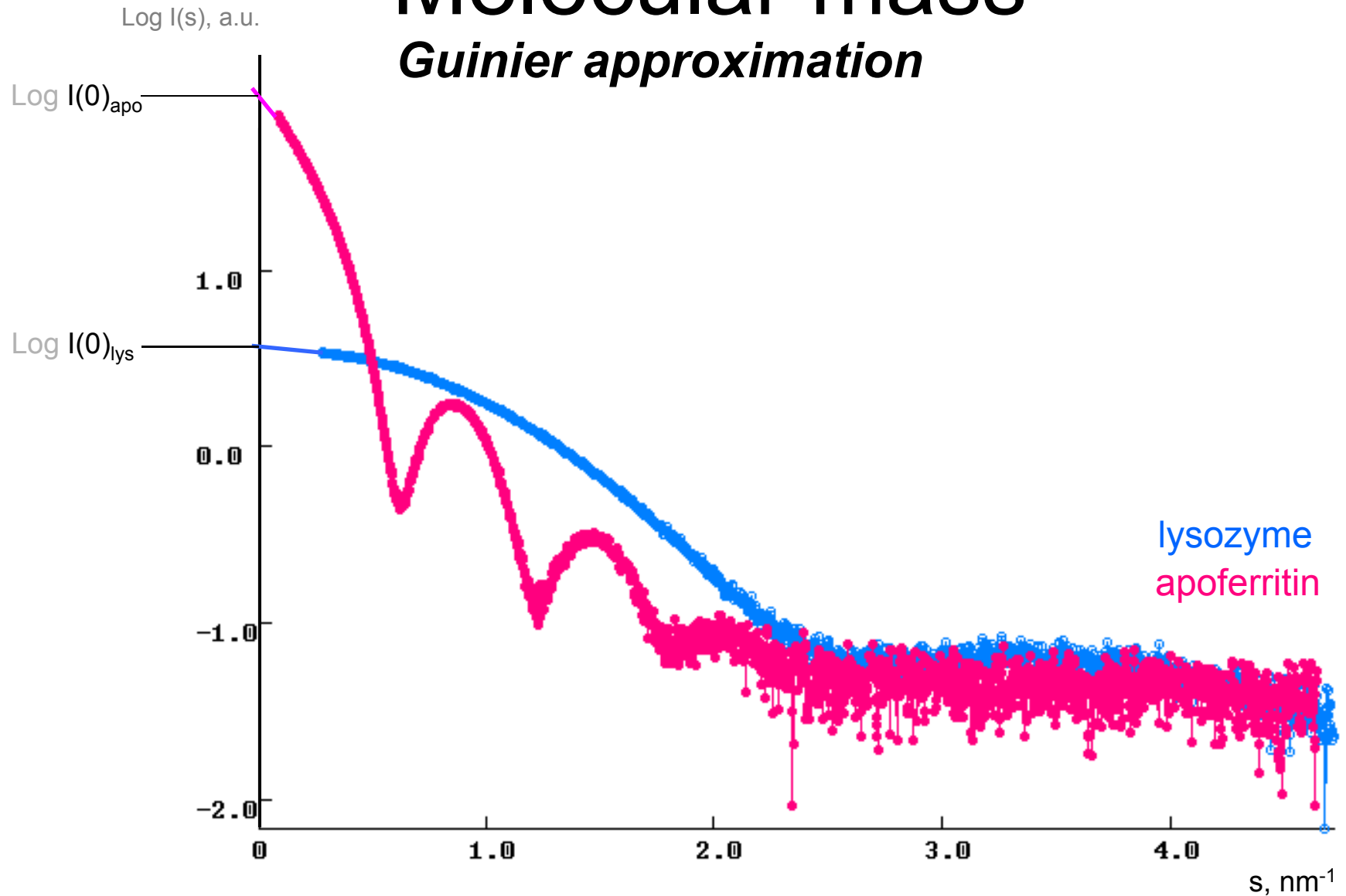
Radius of gyration (R_g)

Guinier plot



Molecular mass

Guinier approximation



$I(0)$ and Molecular Mass

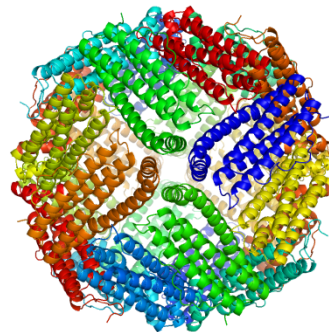
$$\frac{MM_{\text{sample}}}{MM_{\text{BSA}}} = \frac{I(0)_{\text{sample}}}{I(0)_{\text{BSA}}}$$

$$MM_{\text{sample}} = I(0)_{\text{sample}} * MM_{\text{BSA}} / I(0)_{\text{BSA}}$$

BSA
 $R_g = 3.1 \text{ nm}$
 $I(0) = 11.7$
 $MM_{\text{BSA}} = 66 \text{ kDa}$



$R_g = 1.46 \text{ nm}$
 $I(0) = 3.66$
 $MM = 20.6 \text{ kDa}$



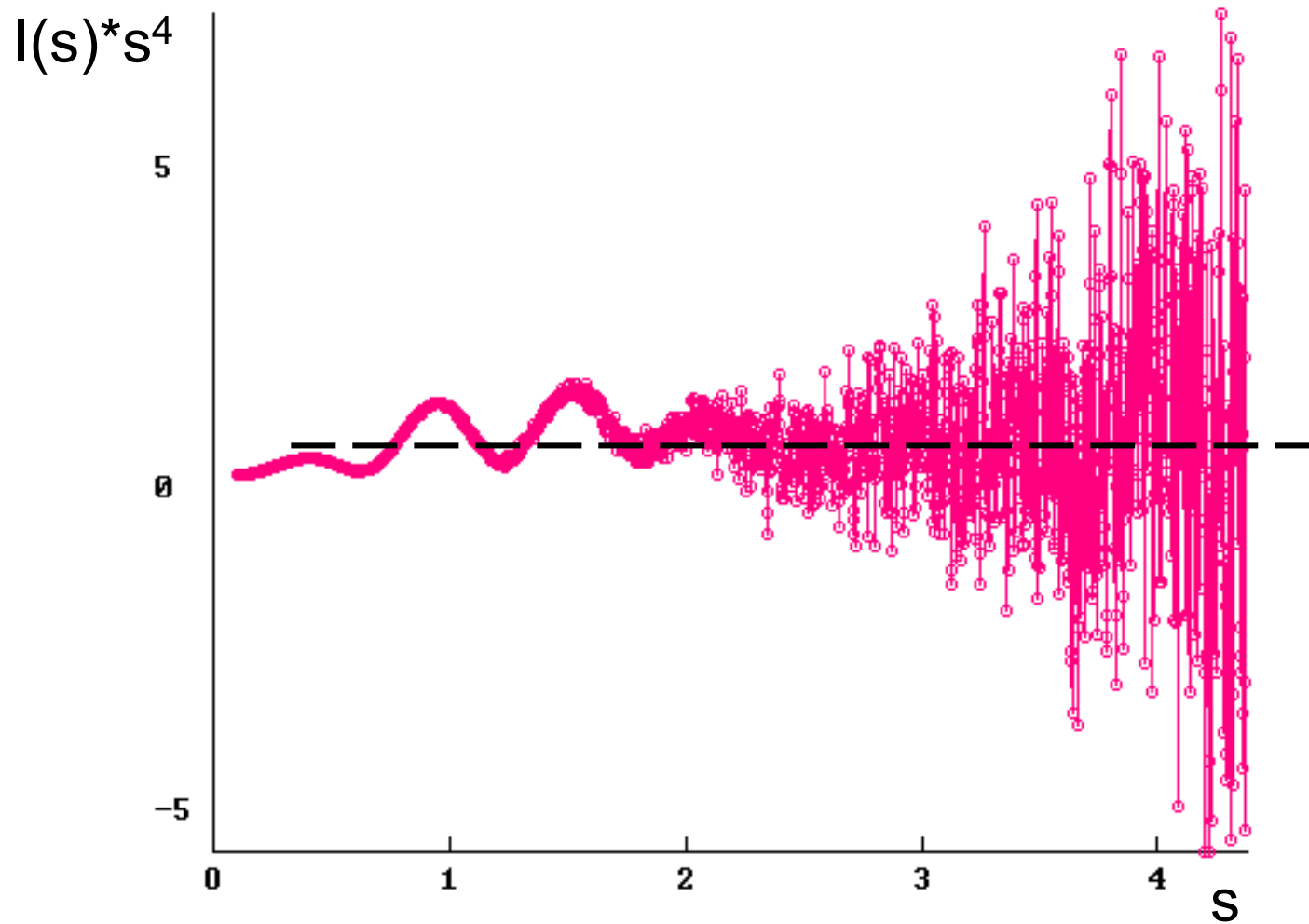
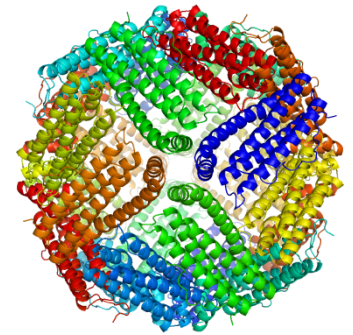
$R_g = 6.81 \text{ nm}$
 $I(0) = 79.45$
 $MM = 448.2 \text{ kDa}$

Porod law

$$I(s) \sim s^{-4}$$

Intensity decay is proportional to s^{-4} at higher angles
(for globular particles of uniform density)

Porod law



Porod law

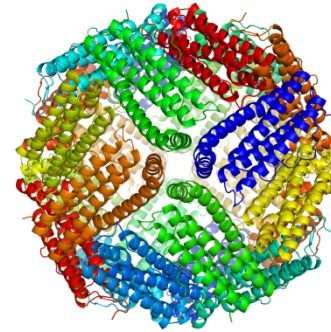
Excluded volume of the hydrated particle

$$V_P = \frac{2\pi^2 I(0)}{\int_0^\infty [I(s) - K_4] s^2 ds}$$

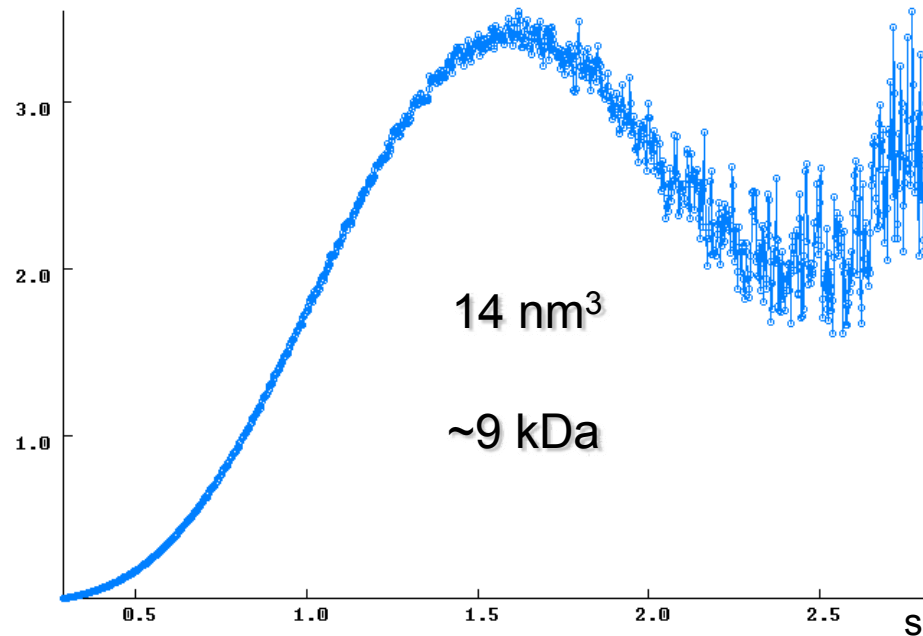
K_4 is a constant determined to ensure the asymptotical intensity decay proportional to s^{-4} at higher angles following the Porod's law for homogeneous particles

Porod plot

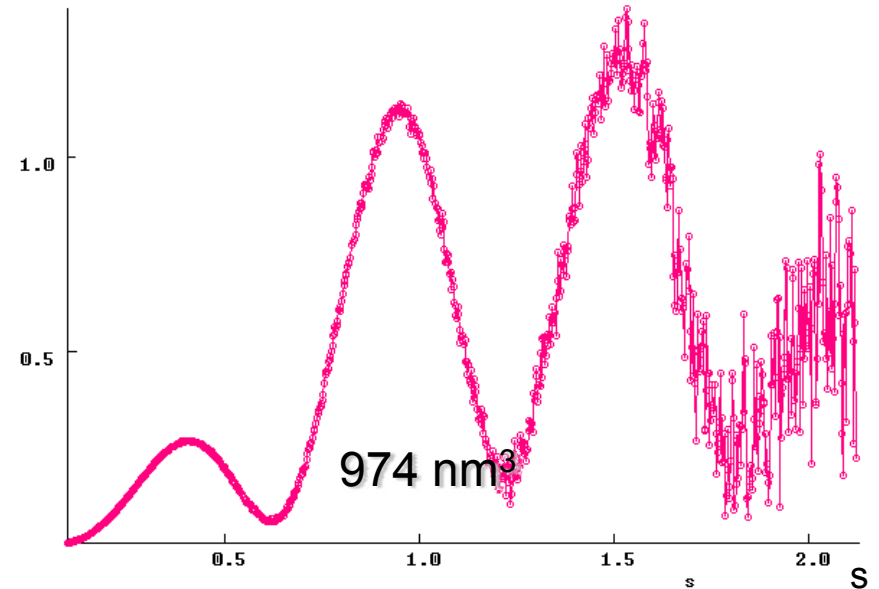
Excluded volume of the hydrated particle



$I(s) \cdot s^4$

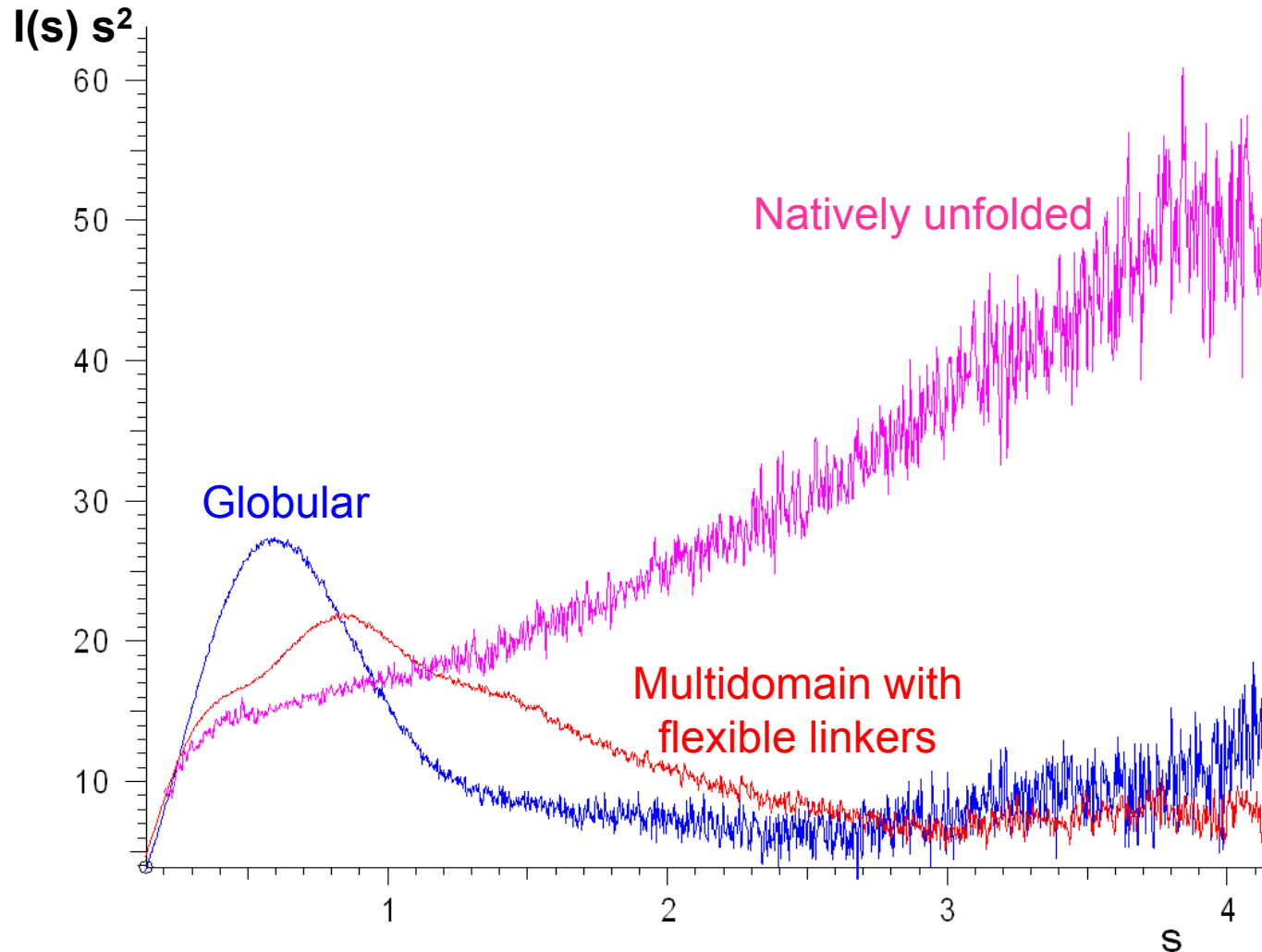


$I(s) \cdot s^4$



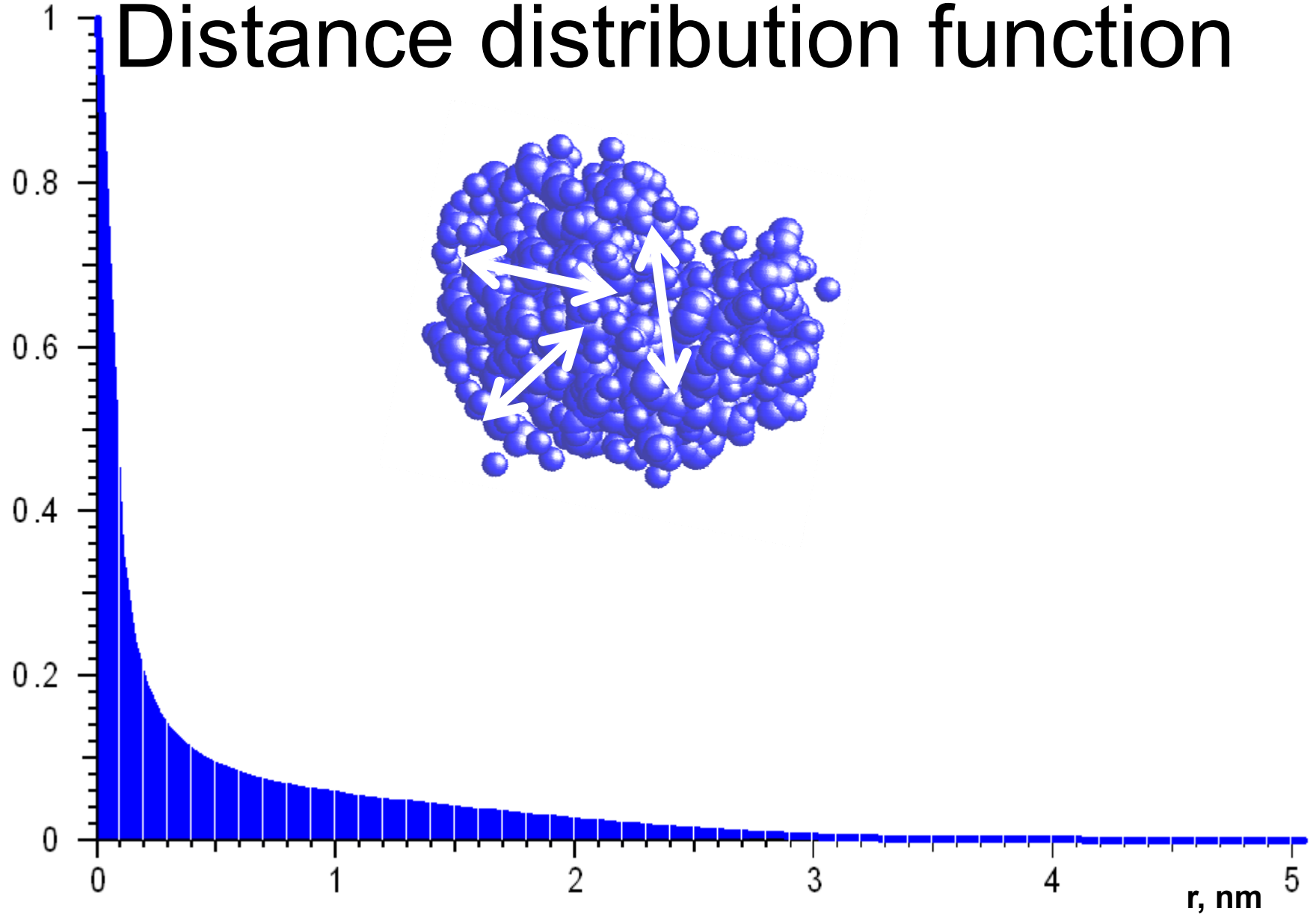
Kratky plot

Patterns of globular and flexible proteins



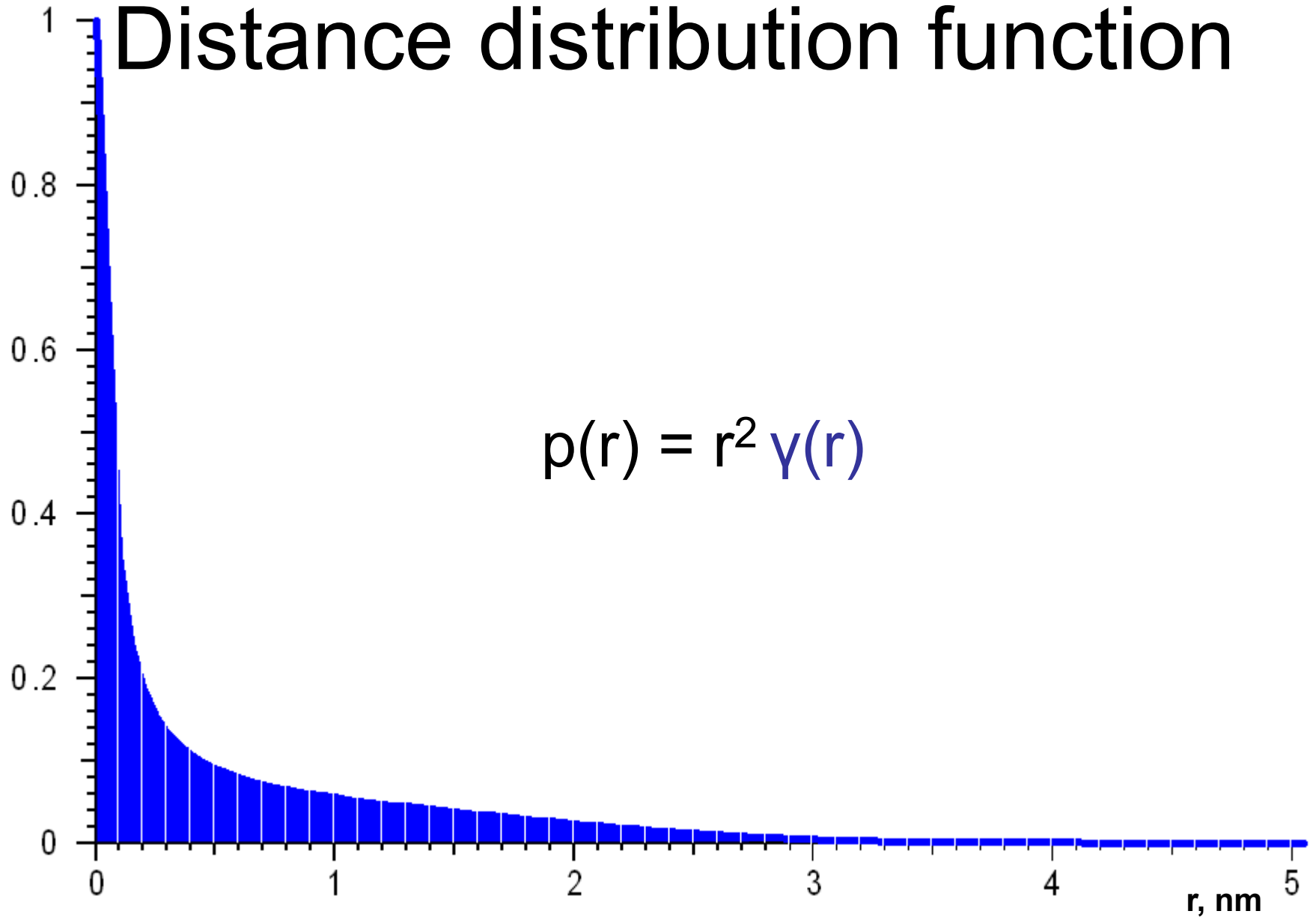
$\gamma(r)$

Distance distribution function

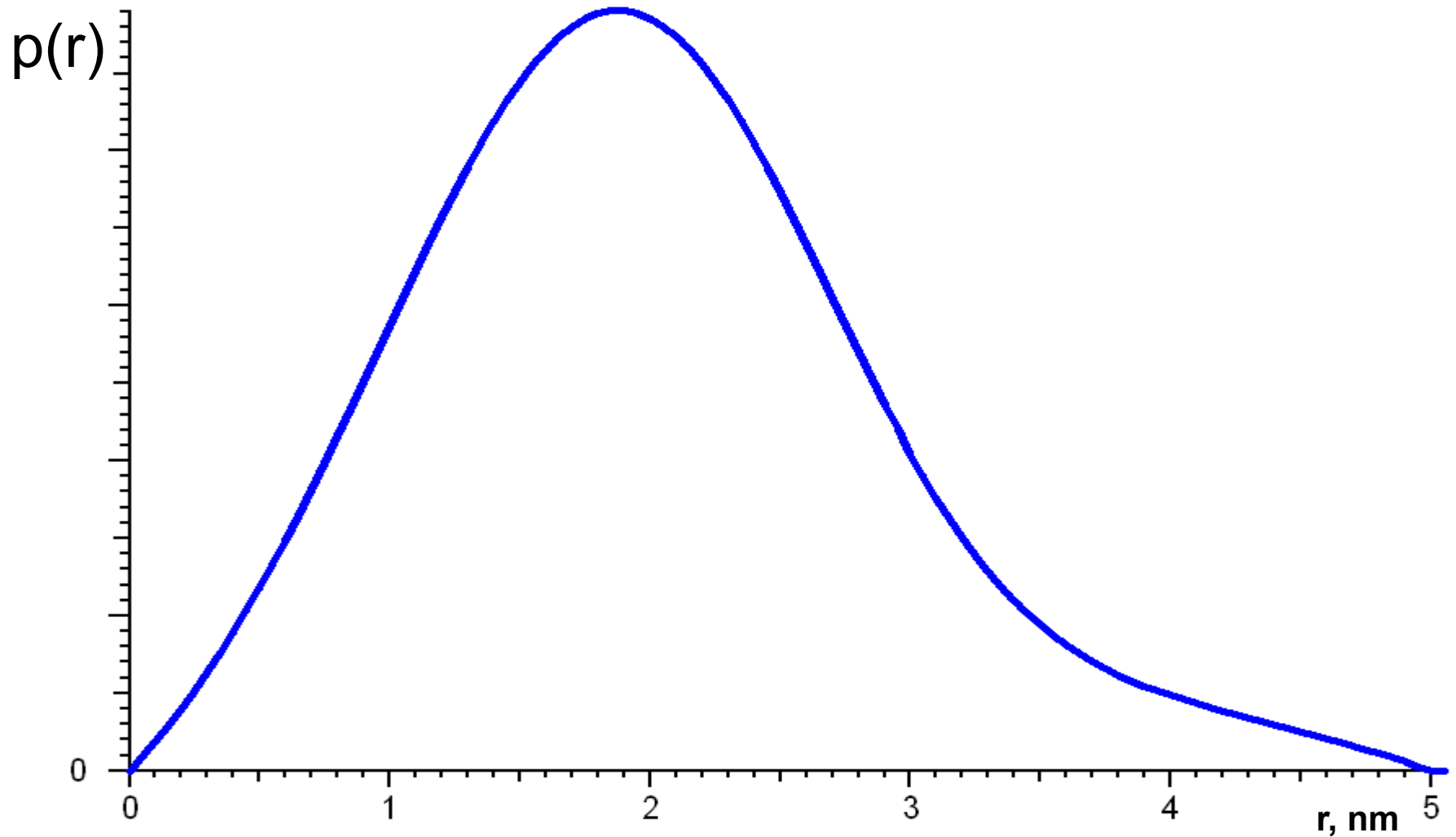


$\gamma(r)$

Distance distribution function

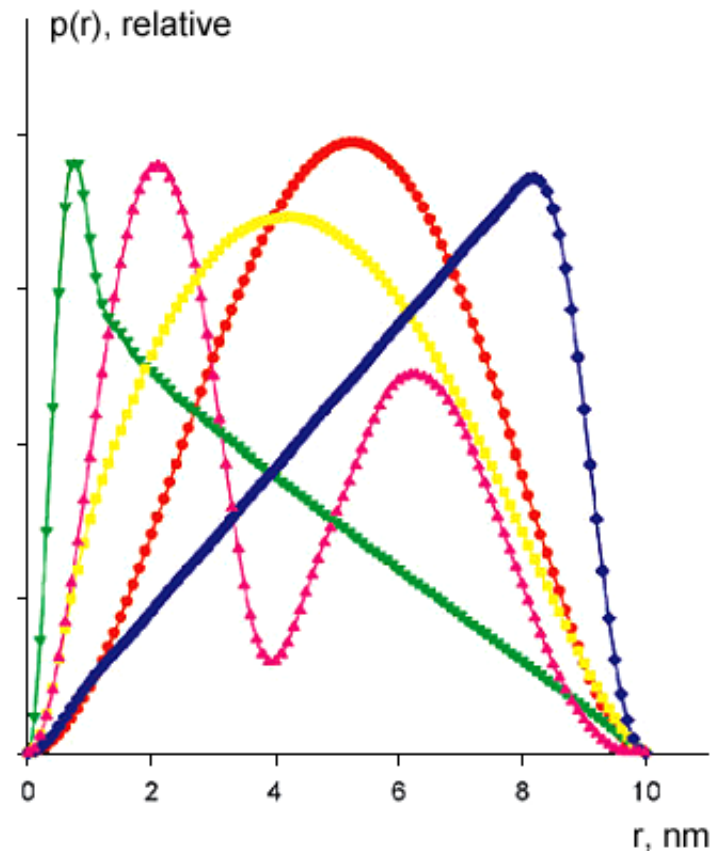
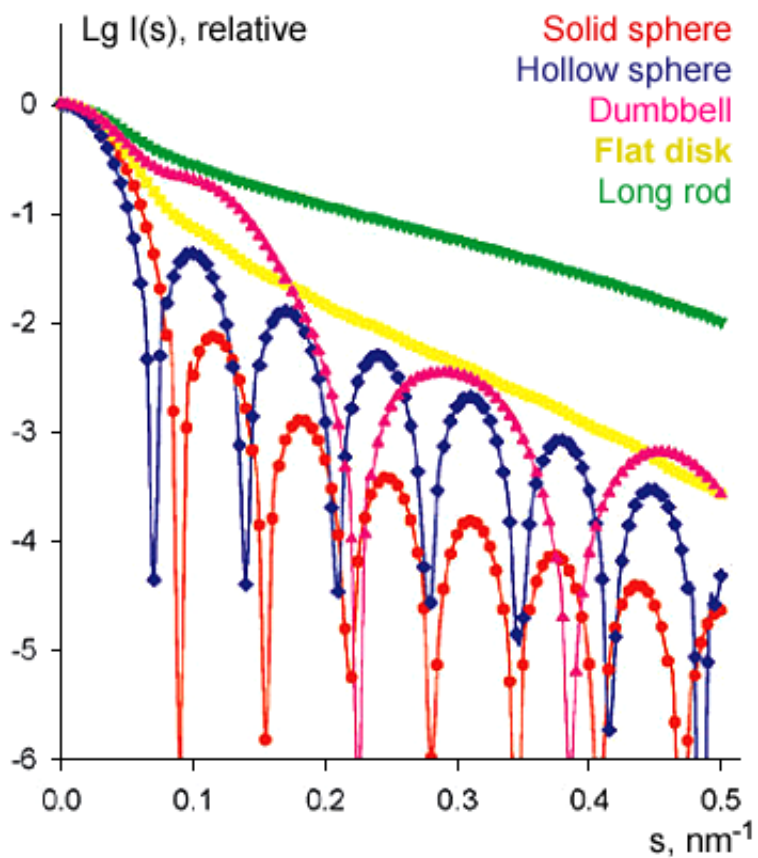


Distance distribution function



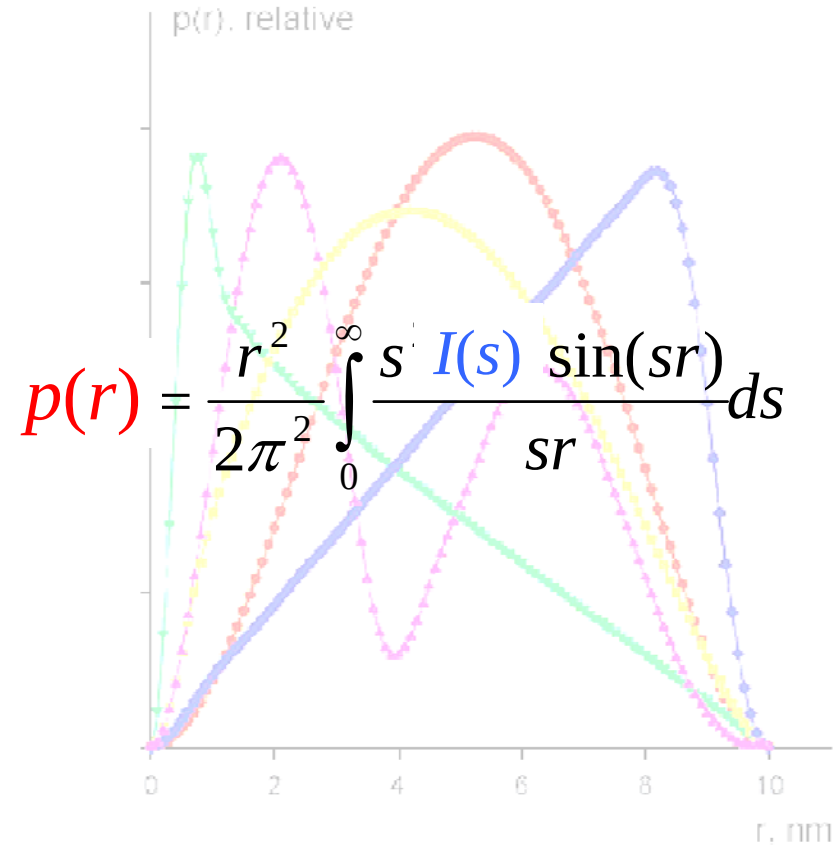
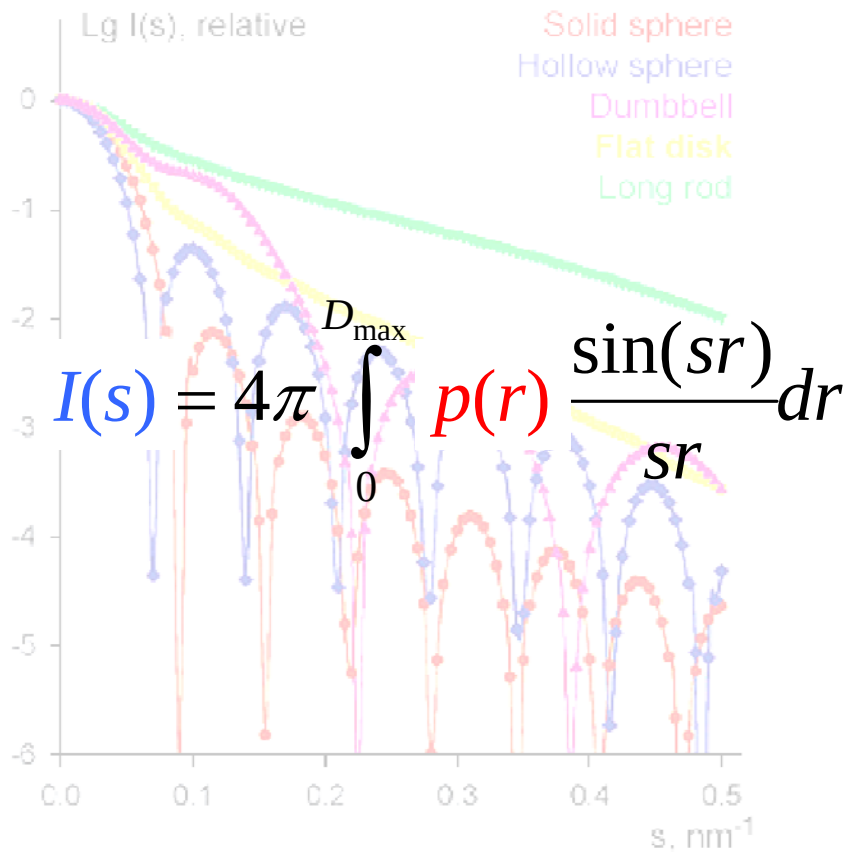
$p(r)$ function

Distance distribution function



$p(r)$ function

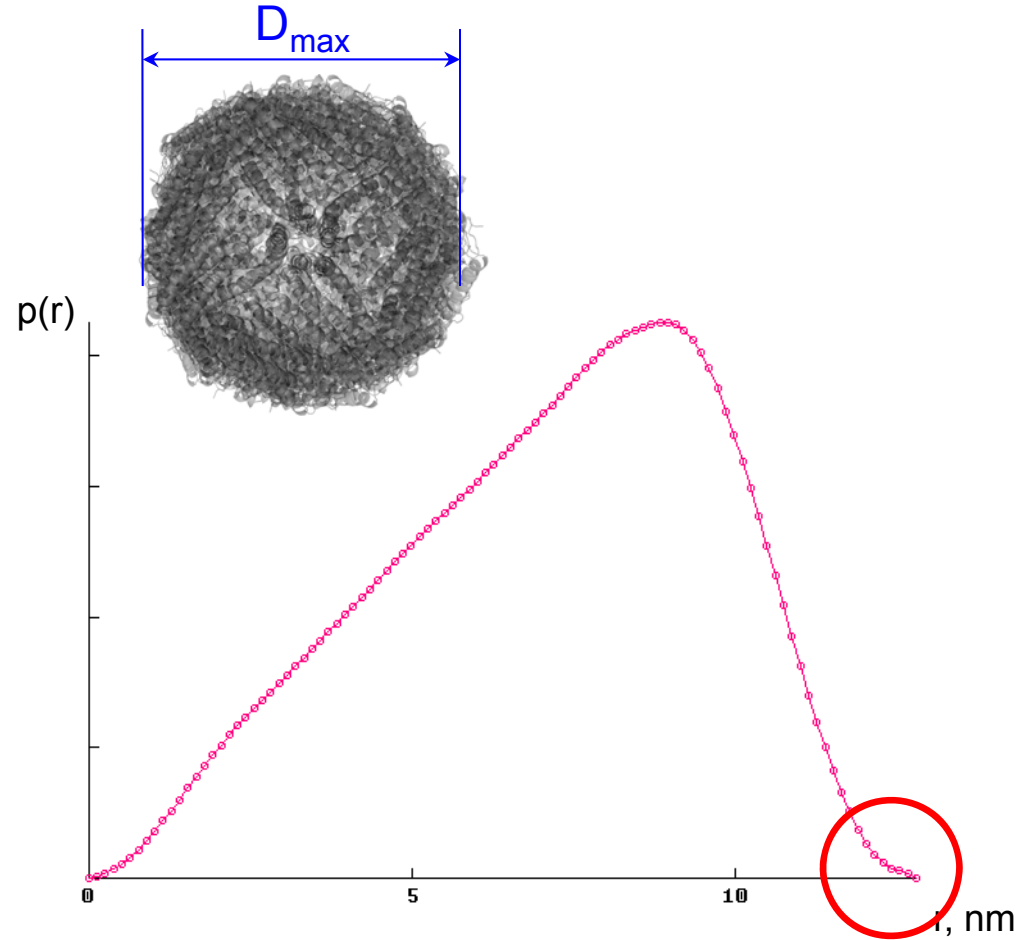
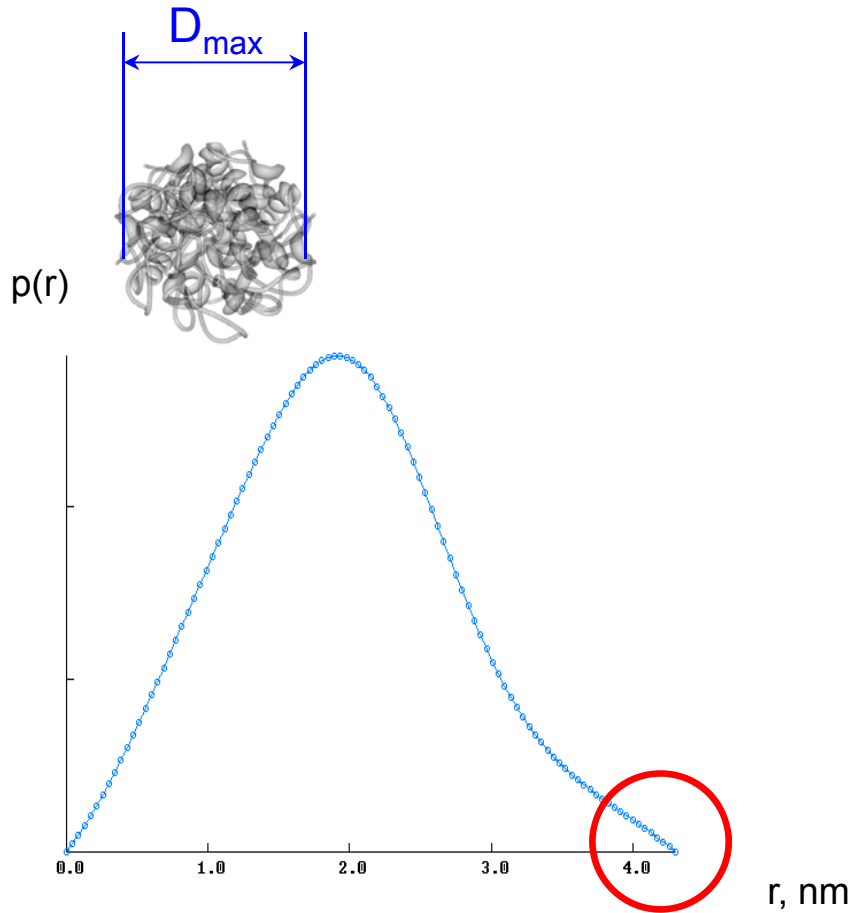
Distance distribution function



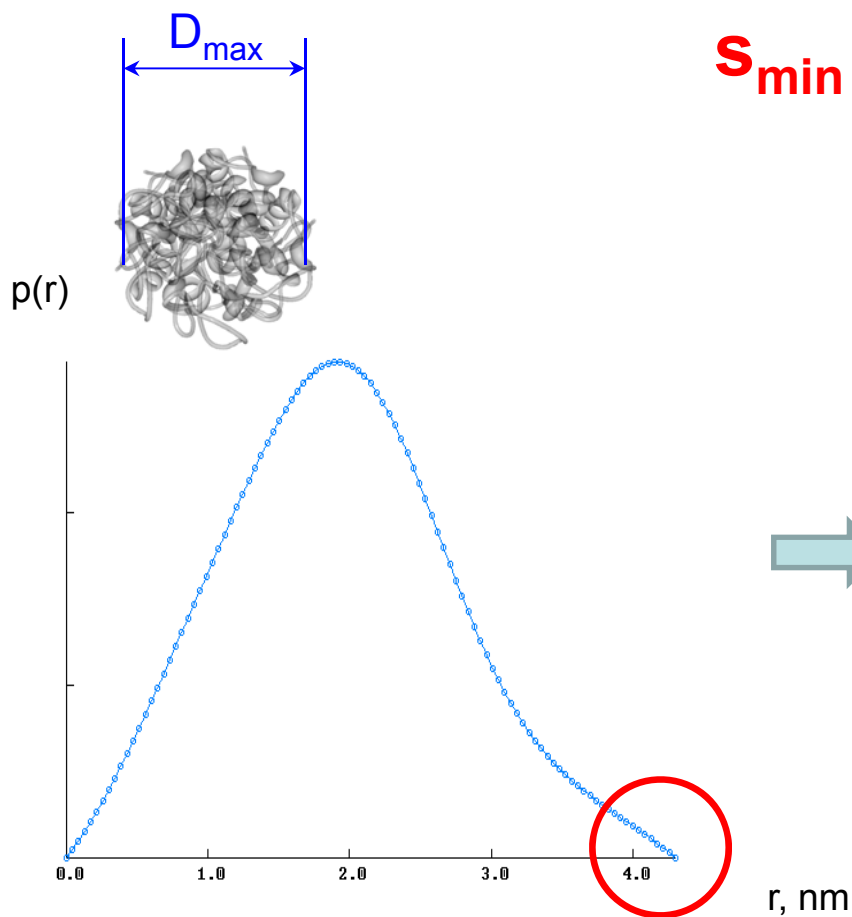
Indirect Fourier Transform

$p(r)$ plot

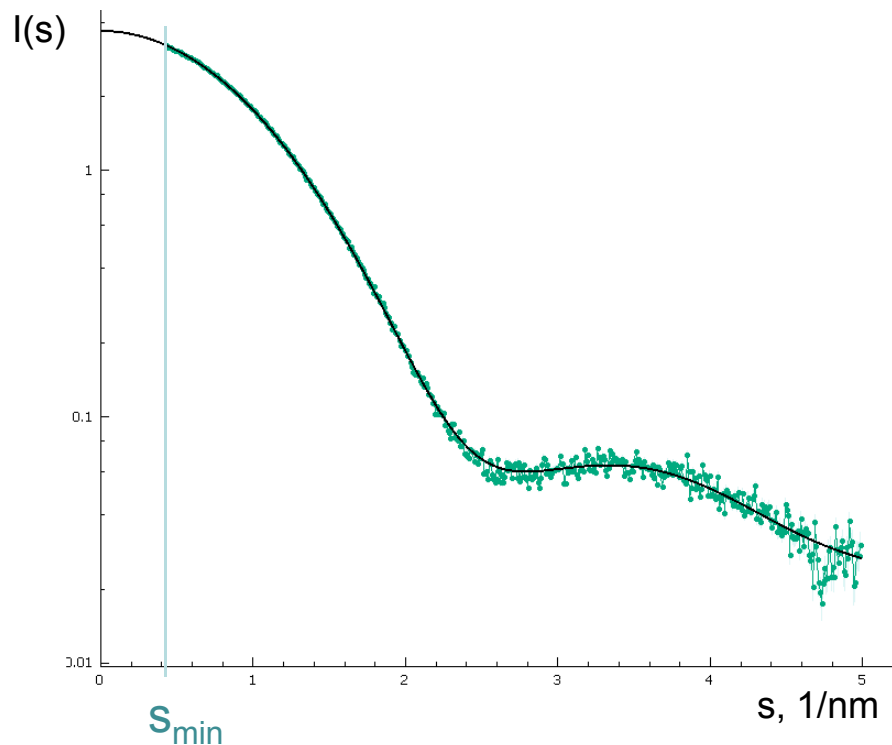
Distance distribution function



Data quality

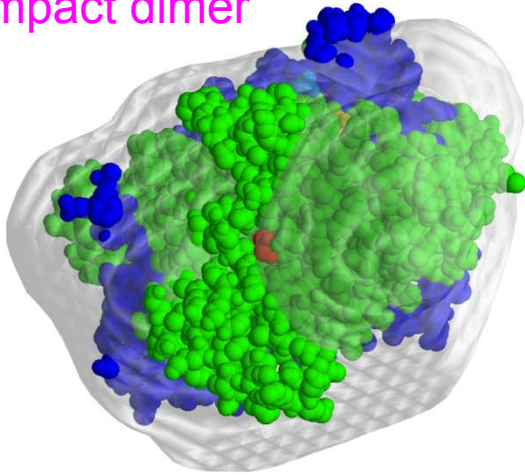


$$s_{\min} \leq \pi/D_{\max}$$

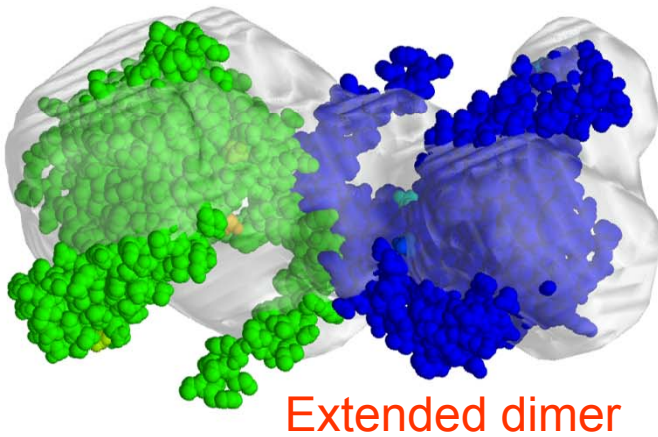
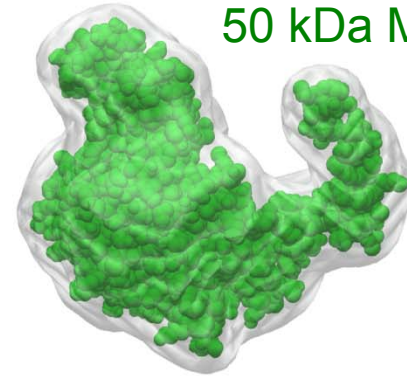


Yeast bleomycin hydrolase 3GCB

Compact dimer

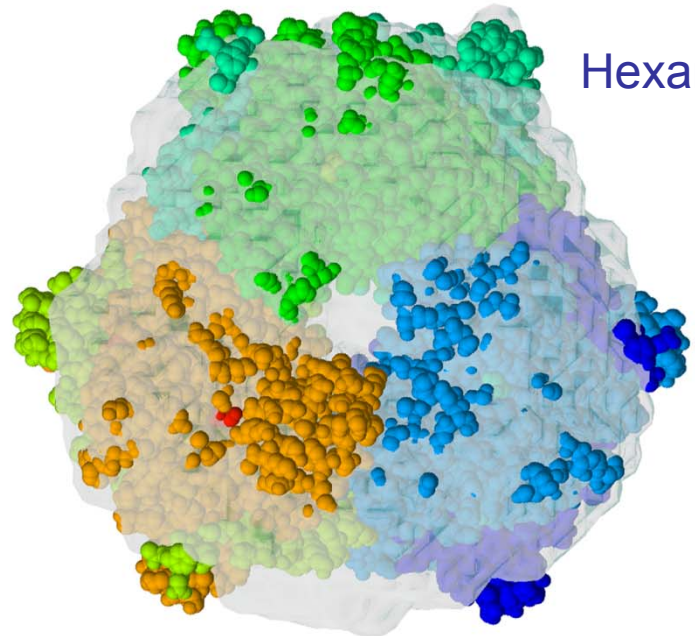


50 kDa Monomer

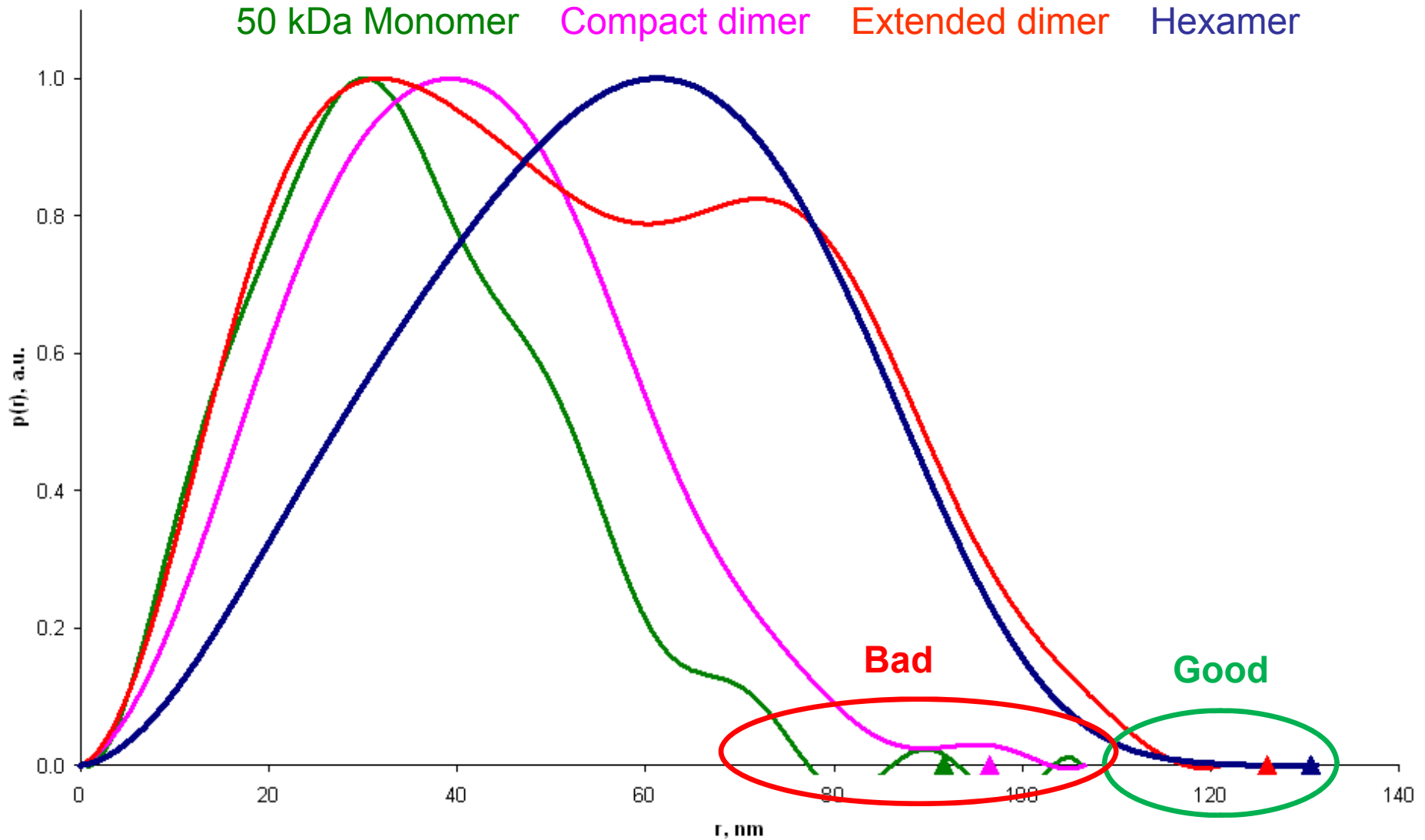


Extended dimer

Hexamer



$p(r)$ plots

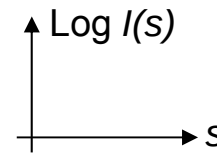


Yeast bleomycin hydrolase 3GCB

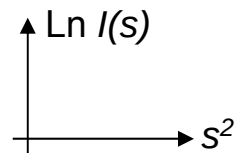
Summary

- Exposure 3D $\rightarrow$ 2D
- Radial averaging $\rightarrow$ 1D
- Normalization
- Background subtraction
- Analysis

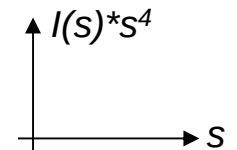
- Log plot



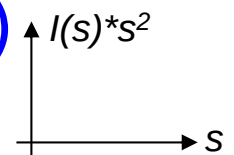
- Guinier plot (R_g , MM)



- Porod plot



- Kratky plot (flexibility)



- $p(r)$ plot



POWERED BY WORLD'S NEW FAVOURITE SUPERHERO

SAXSMAN

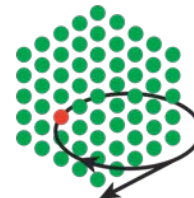
THE INCREDIBLE SMALL-ANGLE X-RAY SOLUTION SCATTERING MAN[®]



Thank you!

After questions: group photo!

BI((O))SAXS



www.saxier.org/forum